Iterative Training Sampling Coupled With Active Learning for Semisupervised Spectral–Spatial Hyperspectral Image Classification

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***Abstract*— Training sample selection is a great challenge for hyperspectral image classification (HSIC), specifically when only a very limited number of labeled data samples are available for training. Two recently developed concepts for training sample selection are of particular interest. One is active learning (AL), which augments labeled training samples by including unlabeled data samples as new training samples. The other is iterative training sampling (ITS), which expands data cubes by including additional spatial classification information into the training samples. This article combines AL and ITS simultaneously to derive a joint ITS–AL spectral–spatial (SS) classification approach, to be called ITS augmentation by AL spectral–spatial classification, referred to as ITSA-AL-SS, which can improve SS classification using either AL or ITS alone. The novel idea of ITSA-AL-SS is to take advantage of ITS to expand data cubes by incorporating additional spatial classification information iteratively into the AL-augmented unlabeled data samples to update the current training samples iteration by iteration in one-shot operation. As expected, ITSA-AL-SS is benefited from both ITS and AL to not only further improve classification accuracy but also reduce classification inconsistency.**

***Index Terms*— Active learning (AL), iterative training sampling (ITS), iterative training sampling augmentation (ITSA), spectral–spatial (SS).**

NOMENCLATURE

AA	Average accuracy.
AL	Active learning.
BT	Breaking ties.
GT	Ground truth.
HSIC	Hyperspectral image classification.
IEPF	Iterative edge preserving filter.
IFTSA-AL-SS	Iterative fixed training sample augmentation by AL SS classification.
IRTSA-AL-SS	Iterative random training sample augmentation by AL SS classification.
ITS	Iterative training sampling.
ITSA	Iterative training sampling augmentation.
ITSA-AL-SS	Iterative training sampling augmentation by AL spectral-spatial classification.
LORSAL	Logistic regression via variable splitting and augmented Lagrangian algorithm.
MAP	Maximum *a posteriori.*
MBT	Modified breaking ties.
MI	Mutual information.
MLL	Multilevel logistic.
MLR	Multinomial logistic regression.
MPM-LBP	Maximizer of the posterior marginal by loopy belief propagation.
OA	Overall accuracy.
RS	Random selection.
RTS	Random training sampling.
SD	Standard deviation.
SF	Spatial filter.
SS	Spectral–spatial.
TI	Tanimoto index.
n_{AL}	Number of unlabeled training samples augmented in each iteration via AL.
$n_{\mathrm{unlabeled}}^{\mathrm{total}(k)}$	Size of $\{\cup_{j=0}^{k} S_{\mathrm{unlabeled}}^{n_{\mathrm{AL}}(j)}\}$, total number of augmented unlabeled training samples at the kth iteration.
$n_{\mathrm{training}}^{(k)}$	Size of $S_{\mathrm{training}}^{(k)}$.
S_{labeled}	Initial labeled sample set.
$S_{\mathrm{labeled}}^{(k)}$	kth labeled sample set.
$S_{\mathrm{training}}^{(k)}$	kth training sample set.
$S_{\mathrm{unlabeled}}^{(k)}$	kth unlabeled sample set.
$S_{\mathrm{unlabeled}}^{n_{\mathrm{AL}}(k)}$	kth informative unlabeled sample set selected by AL.
δ	Spectral-based probabilistic classifier.
Ω_{HSI}	Data set provided by GT$= \{\mathbf{r}_n\}_{n=1}^{N}$.
$\Omega^{(k)}$	Data cube produced at the kth iteration.
$\mathrm{PMap}_m^{(k)}$	mth class probabilistic classification map produced by δ using $S_{\mathrm{training}}^{(k)}$ as training sample set.

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$\text{SFPMap}_m^{(k)}$	mth class spatial filtered probabilistic classification map produced from $\text{PMap}_m^{(k)}$.
$\text{BMap}_m^{(k)}$	mth class binary classification map produced from $\text{SFPMap}_m^{(k)}$ using MAP.

I. Introduction

SUPERVISED HSIC [1]–[13] requires a judicious selection of training samples to perform well. Unfortunately, in many applications acquiring a large number of training samples is challenging due to its expensive cost or practical infeasibility. As a result, training sample selection is one of key factors that have significant impacts on classification performance. Two recently developed approaches are of particular interest.

One is semisupervised classification [14]–[24] which augments selected labeled training samples by including certain specific unlabeled data samples selected by AL or self-learning models from the original data set iteratively. Such AL-SS classification usually begins with a small set of labeled samples as an initial training sample set; it calculates the uncertainty score of each classified unlabeled data sample and then selects a certain number of unlabeled data samples with highest uncertain scores, denoted by n_{AL}, as new training samples to augment the current training sample set iteratively. So, unlike the commonly used RTS which randomly selects training samples according to the GT as *a priori* training samples, the unlabeled data samples selected by AL can be considered as *a posteriori* training samples. Accordingly, the classifier can be trained by high informative and confident training samples to achieve better classification performance, while reducing the cost of training samples' collection.

As an alternative to AL, a second approach, called ITS, was recently developed in [25]–[28]. Instead of augmenting the current training samples as AL does, ITS selects and updates the training samples from an expanded data cube by a feedback loop via an iterative process while keeping the same set of training sample pool during the entire iterations. More specifically, ITS expands the current data cube by including *a posteriori* spatial filtered classification maps to form a new data cube via a feedback loop in an iterative manner and then selects the same set of training samples from the new expanded data cube through an iterative process.

There are four striking similarities between AL and ITS in the sense of seeking different aspects to better improve training sample selection. First, both make attempts to manipulate training sample selection in two completely different concepts. AL augments training sample sizes by including unlabeled data samples as new training samples, while ITS expands data cubes by including new spatial classification information into training samples. Second, both use the GT to randomly select an initial set of training samples as *a priori* training samples. After initialization, AL then works on spectral classification maps to find informative unlabeled data samples as new training samples. Compared with AL, ITS applies SFs to the spectral classification maps to obtain *a posteriori* spatial classification maps to provide new spatial information to be included in selected training samples. Third, AL finds classified unlabeled data samples with high informativeness and confidence to augment the current training samples to create "a new training sample set." In contrast, ITS implements SFs on classification maps to capture *a posteriori* spatial classification maps to be added to the current data cube to create "a new data cube." Finally, AL uses an iterative process to augment the current training samples, whereas ITS expands the current input data cube by ITS via feedback loops.

In spite of the above four similarities, there are also fundamental differences between AL and ITS in terms of concepts and implementations. For example, although both ITS and AL use iterative processes, ITS uses feedback loops to expand data cubes compared with AL which uses certain criteria to find unlabeled data samples with high informativeness to augment training sample sets without feedback. So, ITS keeps the training sample size of each class fixed and unchanged during the entire iterative process, but AL uses the same original data set throughout the entire iterative process. Another is that ITS selects the same set of training samples from expanded data cubes for each iteration, whereas AL keeps adding new training samples to grow the original training sample set after each iteration. A third one is that ITS expands data cubes by adding *a posteriori* spatial filter classification maps via feedback loops iteration by iteration. As a result, the same set of training samples will be updated by extra spatial information provided by the newly added spatial filtered classification maps. In this case, ITS does not expand the training data pool in terms of training sample size as opposed to AL which actually uses sampling scheme to grow the current training sample pool while working on the same data set iteration after iteration.

As described above, AL and ITS are derived from different design rationales and have their own separate advantages to improve classification performance. This article develops an approach to combine these two into one joint technique, called iterative training sample augmentation by AL (ITSA-AL) which explores complementary benefits offered by AL and ITS. The main idea of ITSA-AL is to include training sample augmentation by AL in the iterative feedback process implemented in ITS at each iteration so that the *a posteriori* probability maps resulting from SS classifier not only can be used to augment the current training samples but also can be fed back to expand the current data cube to include new *a posteriori* SS information simultaneously. So, on one hand, ITSA-AL can be considered as an ITS-expanded data cube version of AL. On the other hand, ITSA-AL can also be considered as an AL-augmented training sampling version of ITS. Most importantly, ITSA-AL can complement each other to make up their deficiency. Specifically, ITS usually had weak classification performance when the available training samples are limited. In this case, by including AL-augmented unlabeled data samples, the ITS can start with a very small number of training samples and gradually grow the training sample pools to enhance its classification accuracy.

By incorporating ITS in AL, two significant drawbacks suffered from AL can be remedied. One is the termination of the iterative process carried out by AL. Typically, this is done by predetermining the number of augmented training

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samples, n_{AL} empirically such as $n_{\mathrm{AL}} = 10$ or 15 and setting a maximum number of iterations such as 40 to terminate the AL process. This issue can be resolved by ITS which implements an automatic stopping rule [25]–[28]. The other is the uncertainty introduced by randomness resulting from the augmentation of training samples during the iterative process. The data cubes expanded by ITS include gradually increased *a posteriori* spatial classification information which reduces the misclassification errors, and in the meantime also increases the *a posteriori* probability estimation accuracy. In particular, two versions of ITS recently developed in [28] can be used for this purpose. One is called iterative fixed training sampling (IFTS) which randomly selects an initial training sample set and then updates the same selected set of training samples from an expanded data cube iteratively [27]. In other words, the same initial set of training samples that are randomly selected at the beginning is fixed and remains unchanged during the entire iterative process. But it should be noted that the same training samples are indeed updated from the expanded data cubes iteration by iteration by including more and more spatial classification information captured by its implemented SFs. On the other hand, different from IFTS, the other is called iterative random training sampling (IRTS) developed in [28] which always selects a new random initial set of training samples from an expanded data cube as long as it begins with a new iteration. So, the key difference between IFTS and IRTS lies in the fact that IFTS is a "static" training sampling, while IRTS is a "dynamic" training sampling. These two are further used to extend ITSA-AL into two semisupervised algorithms to be called iterative fixed training sample augmentation by AL (IFTSA-AL) and iterative random training sample augmentation by AL (IRTSA-AL).

Several key contributions of this article can be summarized as follows.

1) The first and the foremost is the joint use of ITS and AL to update and augment an initially selected set of training samples iteratively. It results in two new algorithms, IFTSA-AL-SS and IRTSA-AL-SS.
2) Both ITS and AL use *a posteriori* classification information to update training samples from expanded data cubes by ITS and augment training samples by AL criteria simultaneously. That is, updating and augmenting training samples are carried out in one-shot operation.
3) On one hand, the ITS-selected training samples are augmented by AL to include more unlabeled data samples to improve classification. On the other hand, the AL-augmented training samples are improved by including extra spatial classification information provided by expanded data cubes iteratively.
4) An automatic stopping rule is developed to terminate both ITS and AL processes. This is particularly crucial for AL where there is no such rule available on how to terminate its iterative training sample augmentation process.

The remainder of this article is organized as follows. Section II reviews the process of AL acquiring informative unlabeled samples. Section III describes the details of the proposed IFTSA-AL-SS and IRTSA-AL-SS algorithms. Section IV presents the classification performance measures used for experiments. Sections V and VI conduct experiments along with discussions. Section VII draws some conclusions.

II. AL Training Sample Augmentation

AL has been used to develop semisupervised HSIC by iteratively including unlabeled data samples to augment the current training sample set via some certain informativeness measures [14]–[24]. The idea of AL is to calculate the uncertainty score of each unlabeled classified data sample and then to select classified data samples with highest uncertainty scores up to a number, n_{AL} which is predetermined empirically. These n_{AL} selected unlabeled classified data samples are further included into the current training sample set to form an augmented new training sample set for the next round classification. As a consequence, AL shows significant improvement in classification accuracy by only using a limited number of labeled training samples.

A. Training Sample Augmentation by AL

Generally speaking, there are four stages to implement a general AL-based training sample augmentation. Assume that $\{\mathbf{r}_n\}_{n=1}^{N}$ and $\{\mathbf{r}_i^{\mathrm{training}}\}_{i=1}^{n_{\mathrm{training}}}$ are the set of data samples and a selected set of training samples, respectively, where n_{training} is the number of training samples. In the first stage, AL uses a spectral probabilistic classifier $\delta(\mathbf{r}_n)$ to estimate *a posteriori* probability for each classified data sample $\mathbf{r}_n$ indicating its probability assigned to the mth class, C_m, denoted by $p(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)$. Next, in the second stage, an AL criterion is used to compute the informativeness score for each classified unlabeled data sample $\mathbf{r}_n$. Then, in the following third stage, those unlabeled data samples with highest AL-calculated informativeness scores are selected to form a set of informative unlabeled data samples, denoted by $\{\mathbf{r}_i^{\mathrm{unlabeled}}\}_{i=1}^{n_{\mathrm{unlabeled}}}$, which will be incorporated to augment the current training sample set $\{\mathbf{r}_i^{\mathrm{training}}\}_{i=1}^{n_{\mathrm{training}}}$ to create a new training sample set, $\{\mathbf{r}_j^{\mathrm{augment}}\}_{j=1}^{n_{\mathrm{augment}}} = \{\mathbf{r}_i^{\mathrm{training}}\}_{i=1}^{n_{\mathrm{training}}} \cup \{\mathbf{r}_i^{\mathrm{unlabeled}}\}_{i=1}^{n_{\mathrm{unlabeled}}}$. Finally, in the fourth stage, the new augmented training sample set $\{\mathbf{r}_j^{\mathrm{augment}}\}_{j=1}^{n_{\mathrm{augment}}}$ will be further used to run the classifier $\delta(\mathbf{r}_n)$ on $\{\mathbf{r}_n\}_{n=1}^{N}$ for classification again. The same four-stage process is repeated over and over again until AL achieves a stopping criterion, such as a prescribed maximum number of iterations.

To be more specific, let $\delta(\mathbf{r}_n)$ be a probabilistic classifier using $\{\mathbf{r}_i^{\mathrm{training}}\}_{i=1}^{n_{\mathrm{training}}}$ as training samples to generate *a posteriori* probability for each data sample $\mathbf{r}_n \in S_{\mathrm{unlabeled}} = \Omega_{\mathrm{HSI}} - \{\mathbf{r}_i^{\mathrm{training}}\}_{i=1}^{n_{\mathrm{training}}}$ given by

$$\delta(\mathbf{r}_n) = \mathbf{p}(\delta(\mathbf{r}_n)|\mathbf{r}_n) = \begin{bmatrix} p(\delta(\mathbf{r}_n) = 1|\mathbf{r}_n) \\ p(\delta(\mathbf{r}_n) = 2|\mathbf{r}_n) \\ \vdots \\ p(\delta(\mathbf{r}_n) = M|\mathbf{r}_n) \end{bmatrix} \tag{1}$$

where $\mathbf{p}(\delta(\mathbf{r}_n)|\mathbf{r}_n)$ is an $M \times 1$ column vector. An AL criterion is used to calculate an informativeness score of each unlabeled sample, $\mathbf{r}_n \in S_{\mathrm{unlabeled}}$. Those n_{AL} unlabeled samples with

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Algorithm 1 Training Sample Augmentation by AL

Input: Ω_{HSI}, S_{labeled}, n_{AL}

1) Let $\Omega_{\text{HSI}} = \{\mathbf{r}_n\}_{n=1}^{N}$ be the set of all data samples in a hyperspectral image provided by the ground truth (GT). Also, let $S_{\text{training}}^{(0)} = S_{\text{labeled}}$ be an initial set of labeled samples randomly selected from Ω_{HSI} and $S_{\text{unlabeled}}^{(0)} = \Omega_{\text{HSI}} - S_{\text{training}}^{(0)} = \{\mathbf{r}_n | \mathbf{r}_n \in \Omega_{\text{HSI}} \text{and} \mathbf{r}_n \notin S_{\text{training}}^{(0)}\}$ be considered as the set of unlabeled data samples from Ω_{HSI} but not in $S_{\text{training}}^{(0)}$. Let $k = 0$.
2) At the kth iteration, implement a probabilistic classifier $\delta(\mathbf{r}_n)$ using $S_{\text{training}}^{(k)}$ to produce a set of *a posteriori* probability maps,
$$\left\{\text{PMap}_m^{(k)}\right\}_{m=1}^{M} = \left\{\left\{p^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)\right\}_{n=1}^{N}\right\}_{m=1}^{M}$$
3) Check if a stopping rule is satisfied. If yes, then the algorithm is terminated. Otherwise continue.
4) Use AL to select n_{AL} unlabeled samples with highest informativeness scores to construct an informative unlabeled sample set, $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$, from $S_{\text{unlabeled}}^{(k)}$
$$S_{\text{unlabeled}}^{n_{\text{AL}}(k)} = \text{AL}\left(n_{\text{AL}}, S_{\text{unlabeled}}^{(k)}, \left\{\text{PMap}_m^{(k)}\right\}_{m=1}^{M}\right) \quad (3)$$
5) Incorporate $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$ into $S_{\text{training}}^{(k-1)}$ to form a kth training sample set augmented by
$$S_{\text{training}}^{(k+1)} = S_{\text{traing}}^{(k)} \cup S_{\text{unlabeled}}^{n_{\text{AL}}(k)} = S_{\text{labeled}} \cup \left\{\cup_{j=1}^{k} S_{\text{unlabeled}}^{n_{\text{AL}}(j)}\right\} \quad (4)$$
$$S_{\text{unlabeled}}^{(k+1)} = \Omega_{\text{HSI}} - S_{\text{training}}^{(k+1)} \quad (5)$$
6) Let $k \leftarrow k + 1$ and go to step 2

Output: $\{\text{PMap}_m^{(k)}\}_{m=1}^{M}$

the highest AL scores are selected to form an informative unlabeled sample set $S_{\text{unlabeled}}^{n_{\text{AL}}}$ to create a new training sample set
$$S_{\text{training}}^{\text{new}} = \left\{\mathbf{r}_i^{\text{training}}\right\}_{i=1}^{n_{\text{training}}} \cup S_{\text{unlabeled}}^{n_{\text{AL}}}. \quad (2)$$

An iterative process using (3) is then repeated over again until it meets a stopping rule, such as an upper bound of n_{training}. A detailed step-by-step implementation of the AL-based training sample augmentation is given in Algorithm 1.

Once AL-SS meets a stopping rule, $\{\text{PMap}_m^{(k)}\}_{m=1}^{M}$ will be the final probability maps and further binarized to produce final binary classification maps $\{\text{BMap}_m^{(k)}\}_{m=1}^{M}$ for classification such as MAP.

B. Informativeness Criterion for AL

The informativeness criterion used in step 4 of Algorithm 1 is designed to calculate an informativeness score for each unlabeled data sample which will be used to select appropriate unlabeled data samples as desired training samples for training sample augmentation. Here, the BT criterion developed in [14] is chosen for this purpose due to its effectiveness.

For each $\mathbf{r}_n \in S_{\text{unlabeled}}^{(k)}$, it defines
$$\text{BT}\left(\delta\left(\mathbf{r}_n \in S_{\text{unlabeled}}^{(k)}\right)\right) = \max_{m\in M}\left\{\mathbf{p}^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)\right\} - \max_{m\in(m^{(k)})^c}\left\{\mathbf{p}^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)\right\} \quad (6)$$
where $m^{(k)} = \max_{m\in M}\mathbf{p}^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n \in S_{\text{unlabeled}}^{(k)})$ and $(m^{(k)})^c = \{m\}_{m=1}^{M} - \{m^{(k)}\}$. The BT in (6) provides an uncertainty score indicating the closeness of a data sample to the boundaries between 2 or more classes. Therefore, a smaller value of $\text{BT}(\delta(\mathbf{r}_n))$ has a higher uncertainty contained in sample $\mathbf{r}_n$, in which case $\mathbf{r}_n$ will be selected and included into $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$ to augment the current training sample set for the next round of classification.

According to (6), the first n_{AL} data samples that yield the n_{AL} smallest values of $\text{BT}(\delta(\mathbf{r}_n \in S_{\text{unlabeled}}))$ will be selected to form $S_{\text{unlabeled}}^{n_{\text{AL}}}$. More specifically, assume that $n_{\text{AL}} \leq |S_{\text{unlabeled}}|$ and $\{\text{BT}(\delta(\mathbf{r}_n \in S_{\text{unlabeled}}))\}_{\mathbf{r}_n}$ is rearranged in the ascending order according to
$$\text{BT}(\delta(\mathbf{r}_n \in S_{\text{unlabeled}})) \leq \text{BT}(\delta(\mathbf{r}_{n+1} \in S_{\text{unlabeled}})) \quad (7)$$
for $1 \leq n - 1 \leq |S_{\text{unlabeled}}|$. In this case, $S_{\text{unlabeled}}^{n_{\text{AL}}} = \{\mathbf{r}_n\}_{n=1}^{n_{\text{AL}}}$. Otherwise, $S_{\text{unlabeled}}^{n_{\text{AL}}} = S_{\text{unlabeled}}$ if $|S_{\text{unlabeled}}| < n_{\text{AL}}$.

III. IFTSA/IRTSA-AL-SS Algorithms

Over the past years, many AL-based SS classification (AL-SS) approaches [14]–[24] were proposed to combine the advantages of SS classification with the AL-augmented training samples to improve the classification performance. Even so, they do not take full advantage of information derived from the *a posteriori* classification information. This section presents a novel ITSA-AL-SS approach to address this issue.

The main idea of ITSA-AL-SS is to implement a set of SFs on *a posteriori* probability maps that are used to calculate sample informativeness quantification and then feed the spatial filtered probability maps back to the current image cube to create a new expanded image cube via an iterative process. Such ITSA-AL-SS can be implemented in conjunction with any arbitrary probabilistic spectral classifier along with any type of SFs. So, it can be viewed as a generalization of traditional AL-SS methods and ITS methods. There are several significant advantages resulting from the use of ITSA-AL-SS. One is the use of ITS to capture *a posteriori* spatial classification information provided by the newly AL-augmented training samples so that more *a posteriori* spatial information derived from the new augmented unlabeled data samples can be included into the ITS-expanded image data cubes via feedback loops. Another is that the accuracy of *posteriori* probabilities' estimation using spectral-based probabilistic classifier can be largely improved through the iterative process that gradually incorporates spatial correlation about classified data samples into image cubes to enhance the effectiveness of AL in selecting appropriate training samples. A third one is the use of TI [29] to custom-design an automatic stopping rule to terminate the ITSA-AL-SS. This is a great advantage over many AL-SS which generally needs to determine an

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upper bound on the maximum number of augmented training samples.

In this section, we present two versions of ITSA-AL-SS algorithms, referred to as IFTSA-AL-SS and iterative random training sampling augmentation-based SS by AL (IRTSA-AL-SS), both of which not only augment the training samples by including informative unlabeled samples iteratively but also update the training samples from the expand input image cubes after each iteration. So, to emphasize the role of ITS in training sample selection, ITSA-AL-SS is referred to as IRTSA/IFTSA-AL-SS thereafter in this article.

A. SS Classification Using Iterative Fixed Training Sampling Augmentation by AL (IFTSA-AL-SS)

Assuming that the image cube in the kth iteration is $\Omega^{(k)}$. At each iteration, we first use a spectral classifier to produce kth probability maps, $\{\text{PMap}_m^{(k)}\}_{m=1}^M$, and then implement an SF on $\{\text{PMap}_m^{(k)}\}_{m=1}^M$ to produce a spatial filtered probability maps, $\{\text{SFPMap}_m^{(k)}\}_{m=1}^M$. Then AL would be used to extract informative unlabeled data samples, $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$, using (3)–(5). Then we can add $\{\text{SFPMap}_m^{(k)}\}_{m=1}^M$ to augment the current image cube $\Omega^{(k)}$ by

$$\Omega^{(k+1)} = \Omega^{(k)} \cup \left\{\text{SFPMap}_m^{(k)}\right\}_{m=1}^M \tag{8}$$

then $S_{\text{training}}^{(k)}$ and $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$ can be regenerated via (3)–(5) from the expanded image cube $\Omega^{(k+1)}$.

There are two crucial issues arising in implementing IFTSA-AL. First, when shall the feedback process in (8) start? It should be noted that a feedback process should not begin from the initial stage (i.e., $k = 0$). That is, if only very few labeled samples are used as initial training samples, the resulting classification will certainly produce not good results because augmenting the probability maps would contain too much incorrect information which degrade the classification performance. Second, when shall a feedback process end? Fortunately, a criterion using TI defined as

$$\text{TI}_m^{(k)} = \frac{|\text{Bmap}_m^{(k)} \cap \text{Bmap}_m^{(k-1)}|}{|\text{Bmap}_m^{(k)} \cup \text{Bmap}_m^{(k-1)}|} > \varepsilon \text{ for } 1 \le m \le M \tag{9}$$

can be used to solve this problem where $\text{TI}^{(k)}$ is a similarity measurement of classification maps between two consecutive iterations, and $\{\text{BMap}_m^{(k)}\}_{m=1}^M$ in (10) is a set of M binary classification maps produced by applying MAP [8], [28] to $\{\text{SFPMap}_m^{(k)}\}_{m=1}^M$ defined as

$$\text{BMap}_m^{(k)}(\mathbf{r}_n) = \begin{cases} 1; & \text{SFPMap}_m^{(k)}(\mathbf{r}_n) \\ & > \text{SFPMap}_{j\in\{1,\ldots,M\}, j\neq m}^{(k)}(\mathbf{r}_n) \\ 0; & \text{otherwise} \end{cases}$$
$$\text{BMap}_m^{(k)} = \left\{\text{BMap}_m^{(k)}(\mathbf{r}_n)\right\}_{\mathbf{r}_n\in\Omega^{(k)}}. \tag{10}$$

A large value of TI indicates no significant change between two classification results. So, the algorithm can be terminated. To address the above issues, two thresholds of TI are used with the first threshold ε_1 to determine when the algorithm should be terminated and the second threshold ε_2 to determine when the algorithm should start ITS. In this case, ε_1 is greater than ε_2. If the algorithm is terminated at the kth iteration, then $\{\text{BMap}_m^{(k)}\}_{m=1}^M$ are the final classification results. The details of IFTSA-AL-SS are provided in Algorithm 2.

Algorithm 2 IFTSA-AL-SS

Input: Ω_{HSI}, S_{labeled}, n_{AL}, ε_1, ε_2

1) Let $\Omega_{\text{HSI}} = \{\mathbf{r}_n\}_{n=1}^N = \Omega^{(0)}$ be the set of all data samples in a hyperspectral image provided by the ground truth (GT). Also, let $S_{\text{training}}^{(0)} = S_{\text{labeled}}$ be an initial training set of randomly selected from Ω_{HSI} and $S_{\text{unlabeled}}^{(0)} = \Omega^{(0)} - S_{\text{training}}^{(0)}$ be considered as the set of unlabeled data samples from $\Omega^{(0)}$ but not in $S_{\text{training}}^{(0)}$. Let $k = 0$.
2) At the kth iteration, use a probabilistic classifier $\delta(\mathbf{r}_n)$ with $S_{\text{training}}^{(k)}$ to produce a set of *Ma posteriori* probability maps,
$$\{\text{PMap}_m^{(k)}\}_{m=1}^M = \{\{p^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)\}_{n=1}^N\}_{m=1}^M.$$
3) Implement M spatial filters on $\{\text{PMap}_m^{(k)}\}_{m=1}^M$ to produce the kth spatial filtered probability maps, $\{\text{SFPMap}_m^{(k)}\}_{m=1}^M$
4) Use MAP to produce binary classification maps from $\{\text{SFPMap}_m^{(k)}\}_{m=1}^M$
$$\left\{\text{BMap}_m^{(k)}\right\}_{m=1}^M = \text{MAP}\left(\left\{\text{SFPMap}_m^{(k)}\right\}_{m=1}^M\right).$$
5) Check if TI $> \varepsilon_1$, if yes, then algorithm is terminated. Otherwise, continue.
6) Check if TI $> \varepsilon_2$, if yes, let $\Omega^{(k+1)} = \Omega^{(k)} \cup \{\text{SFPMap}_m^{(k)}\}_{m=1}^M$ and regenerate $S_{\text{training}}^{(k)}$ and $S_{\text{unlabeled}}^{(k)}$ from new image cube $\Omega^{(k)}$. Otherwise let $\Omega^{(k+1)} = \Omega^{(k)}$ and continue.
7) Use AL to select n_{AL} informative unlabeled samples form an n_{AL}-element set, $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$, from $S_{\text{unlabeled}}^{(k)}$
$$S_{\text{unlabeled}}^{n_{\text{AL}}(k)} = \text{AL}\left(n_{\text{AL}}, S_{\text{unlabeled}}^{(k)}, \left\{\text{PMap}_m^{(k)}\right\}_{m=1}^M\right)$$
8) Add $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$ to augment $S_{\text{training}}^{(k)}$ to a new training sample set and generate new unlabeled sample set by
$$S_{\text{training}}^{(k+1)} = S_{\text{training}}^{(k)} \cup S_{\text{unlabeled}}^{n_{\text{AL}}(k)} = S_{\text{labeled}} \cup \left\{\cup_{j=0}^k S_{\text{unlabeled}}^{n_{\text{AL}}(j)}\right\} \tag{11}$$
$$S_{\text{unlabeled}}^{(k+1)} = \Omega^{(k+1)} - S_{\text{training}}^{(k+1)} = S_{\text{unlabeled}}^{(k)} - S_{\text{unlabeled}}^{n_{\text{AL}}(k)} \tag{12}$$
9) Let $k \leftarrow k + 1$ and go to step 2.

Output: $\{\text{BMap}_m^{(k)}\}_{m=1}^M$

B. SS Classification Using Iterative Random Training Sampling Augmentation by AL (IRTSA-AL-SS)

One major drawback of IFTSA-AL-SS is the feedback process suffering from an inherent uncertainty in terms of final classification results caused by the augmented training samples. This phenomenon mainly results from the random initialization of labeled samples, S_{labeled}, because different

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sets of randomly selected S_{labeled} would result in different classification results, and on many occasions such inconstancy can be significantly large. To mitigate this problem, a random version of IFTSA-AL-SS, IRTSA-AL-SS is further developed in the following.

The key difference between IRTSA-AL-SS and IFTSA-AL-SS is that the initial labeled sample set $S_{\text{labeld}}^{(k)}$ at each iteration is randomly selected from $\mathbf{\Omega}^{(k)}$ rather than provided *a priori*. In this case, S_{labeled} in (11) is replaced by a randomly generated training sample set $S_{\text{labeled}}^{(k)}$ and (11) becomes

$$S_{\text{training}}^{(k+1)} = S_{\text{labeled}}^{(k)} \cup \left\{ \cup_{j=0}^{k} S_{\text{unlabeled}}^{n_{\text{AL}}(j)} \right\} \tag{13}$$

where IRTSA-AL-SS not only updates $\{\cup_{j=1}^{k} S_{\text{unlabeled}}^{n_{\text{AL}}(j)}\}$ but also randomly regenerates $S_{\text{labeld}}^{(k)}$ in each iteration. The details of IRTSA-AL-SS are provided in Algorithm 3.

It should also be noted that the above two semisupervised approaches do not specify a particular probabilistic SS classifier which can be any arbitrary spectral classifier coupled with any type of SFs. Figs. 1 and 2 depict the graphic diagrams of these two above algorithms, IFTSA-AL-SS and IRTSA-AL-SS, respectively.

C. Spectral and Spatial A Posteriori Probabilities Estimation

Even though IFTSA/IRTSA-AL-SS can be implemented with any arbitrary probabilistic classifier and followed by any type of SFs for various practical applications, the MLR developed in [30] was particularly selected as a desired probabilistic classifier to calculate the *a posteriori* probability $p(\delta(\mathbf{r}_n) = j|\mathbf{r}_n)$ for each data sample $\mathbf{r}_n$ because MLR has been widely used reported in the AL-SS literatures [15]–[18].

As for SFs, guided filters [31] were selected with the first principal component of data cube as a guidance image, R, to encode the spatial information into the probability maps produced by MLR. The idea of using guided filters for SS classification was originally derived from EPF classification [8] which has shown to be very promising compared with other existing SS classifiers. However, the original version of EPF used a binary support vector machine (SVM) to produce binary maps which cannot be used as *a posteriori* class probability distribution. To overcome this problem, IFTSA-AL-SS and IRTSA-AL-SS use MLR instead of SVM to construct initial spectral classification probability maps, and then follow up with guided filters to capture spatial information from spectral classification results to better estimate *a posteriori* class probabilities.

1) Multinomial Logistic Regression: MLR developed in [30] is used to estimate the probability of sample $\mathbf{r}_n$ assigned to the mth class, C_m, by

$$p\left(\delta^{\text{MLR}}(\mathbf{r}_n) = m|\mathbf{r}_n, \mathbf{W}\right) = \frac{\exp\left(\mathbf{w}_m^T h(\mathbf{r}_n)\right)}{\sum_{j=1}^{M} \exp\left(\mathbf{w}_j^T h(\mathbf{r}_n)\right)} \tag{14}$$

where h is an activation function of the data sample $\mathbf{r}_n$ commonly used in neural networks. It is worth noting that (14) is indeed the well-known softmax function which normalizes an M dimension input vector into a probability distribution consisting of M probabilities. In (12), $h(\mathbf{r}_n)$ is chosen to be the Gaussian radial basis function (RBF) [32]. Specifically, $h(\mathbf{r}_n) = (1, K(\mathbf{r}_n, \mathbf{r}_1^{\text{training}}), \ldots, K(\mathbf{r}_n, \mathbf{r}_{n_{\text{training}}}^{\text{training}}))^T$ with $K(\mathbf{r}_n, \mathbf{r}_j^{\text{training}}) = \exp(-||\mathbf{r}_n - \mathbf{r}_j^{\text{training}}||^2/2\sigma^2)$ being an RBF kernel that only depends on the training samples $\{\mathbf{r}_j^{\text{training}}\}_{j=1}^{n_{\text{training}}}$ and $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_M\}$ is a set of M regression vectors, also known as logistic regressors where $\mathbf{w}_m$ indicates the regressor for C_m and $\mathbf{w}_m^T h(\mathbf{r}_n)$ can be considered as an output score of the data sample $\mathbf{r}_n$ assigned to C_m. Therefore, higher output score indicates more probability of sample $\mathbf{r}_n$ be assigned to C_m. In general, $\{\mathbf{w}_m\}_{m=1}^{M}$ are computed using training samples

Algorithm 3 IRTSA-AL-SS

Input: $\mathbf{\Omega}_{\text{HSI}}, n_{\text{AL}}, S_{\text{labeled}}^{(0)}, \varepsilon_1, \varepsilon_2$

1) Let $\mathbf{\Omega}_{\text{HSI}} = \{\mathbf{r}_n\}_{n=1}^{N} = \mathbf{\Omega}^{(0)}$ be the set of all data samples in a hyperspectral image provided by the ground truth (GT). Also, let $S_{\text{training}}^{(0)} = S_{\text{labeled}}^{(0)}$ be an initial training set that randomly selected from $\mathbf{\Omega}_{\text{HSI}}$ and $S_{\text{unlabeled}}^{(0)} = \mathbf{\Omega}^{(0)} - S_{\text{training}}^{(0)}$ is the set of unlabeled data samples from $\mathbf{\Omega}^{(0)}$ but not in $S_{\text{training}}^{(0)}$. Let $k = 0$.
2) At kth iteration, use probabilistic spectral classifier $\delta(\mathbf{r}_n)$ to produce a set of *Ma posteriori* probability maps,
$$\left\{\text{PMap}_m^{(k)}\right\}_{m=1}^{M} = \left\{ \left\{ p^{(k)}(\delta(\mathbf{r}_n) = m|\mathbf{r}_n) \right\}_{n=1}^{N} \right\}_{m=1}^{M}$$
3) Use M spatial filters on $\{\text{PMap}_m^{(k)}\}_{m=1}^{M}$ to produce kth spatial filtered probability maps, $\{\text{SFPMap}_m^{(k)}\}_{m=1}^{M}$.
4) Use MAP to produce binary classification maps from $\{\text{SFPMap}_m^{(k)}\}_{m=1}^{M}$
$$\left\{\text{BMap}_m^{(k)}\right\}_{m=1}^{M} = \text{MAP}\left(\left\{\text{PMap}_m^{(k)}\right\}_{m=1}^{M}\right)$$
5) Check if TI $> \varepsilon_1$, if yes, the algorithm is terminated. Otherwise continue
6) Check if TI $> \varepsilon_2$, if yes, let $\mathbf{\Omega}^{(k+1)} = \mathbf{\Omega}^{(k)} \cup \{\text{SFPMap}_m^{(k)}\}_{m=1}^{M}$ and regenerate $S_{\text{unlabeled}}^{(k)}$ and $S_{\text{training}}^{(k)}$ from new image cube$\mathbf{\Omega}^{(k+1)}$. Otherwise let $\mathbf{\Omega}^{(k+1)} = \mathbf{\Omega}^{(k)}$ and continue.
7) Use AL to select n_{AL} informative unlabeled samples to form an n_{AL}-element set, $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$, from $S_{\text{unlabeled}}^{(k)}$
$$S_{\text{unlabeled}}^{n_{\text{AL}}(k)} = \text{AL}\left(n_{\text{AL}}, S_{\text{unlabeled}}^{(k)}, \left\{\text{PMap}_m^{(k)}\right\}_{m=1}^{M}\right)$$
8) Randomly generate a new labeled sample set, $S_{\text{labeled}}^{(k+1)}$. Add $S_{\text{unlabeled}}^{n_{\text{AL}}(k)}$ and $S_{\text{labeled}}^{(k)}$ to augment $S_{\text{training}}^{(k)}$ to generate a new training sample set with new unlabeled sample set obtained by
$$\begin{aligned} S_{\text{training}}^{(k+1)} &= \left((S_{\text{training}}^{(k)} \cup S_{\text{unlabeled}}^{n_{\text{AL}}(k)}) - S_{\text{labeled}}^{(k)} \right) \cup S_{\text{labeled}}^{(k+1)} \\ &= S_{\text{labeled}}^{(k+1)} \cup \left\{ \cup_{j=0}^{k} S_{\text{unlabeled}}^{n_{\text{AL}}(j)} \right\} \\ S_{\text{unlabeled}}^{(k+1)} &= \mathbf{\Omega}^{(k+1)} - S_{\text{training}}^{(k+1)} \end{aligned}$$
9) Let $k \leftarrow k + 1$ and go to step 2

Output: $\{\text{BMap}_m^{(k)}\}_{m=1}^{M}$

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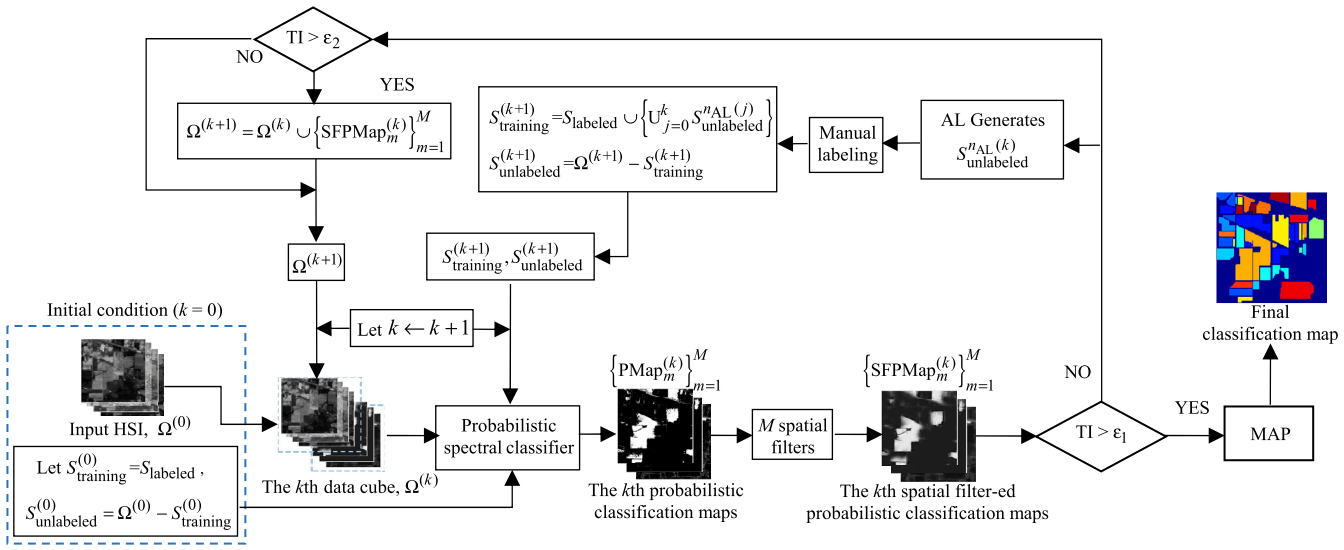


Fig. 1. Flowchart of IFTSA-AL-SS.

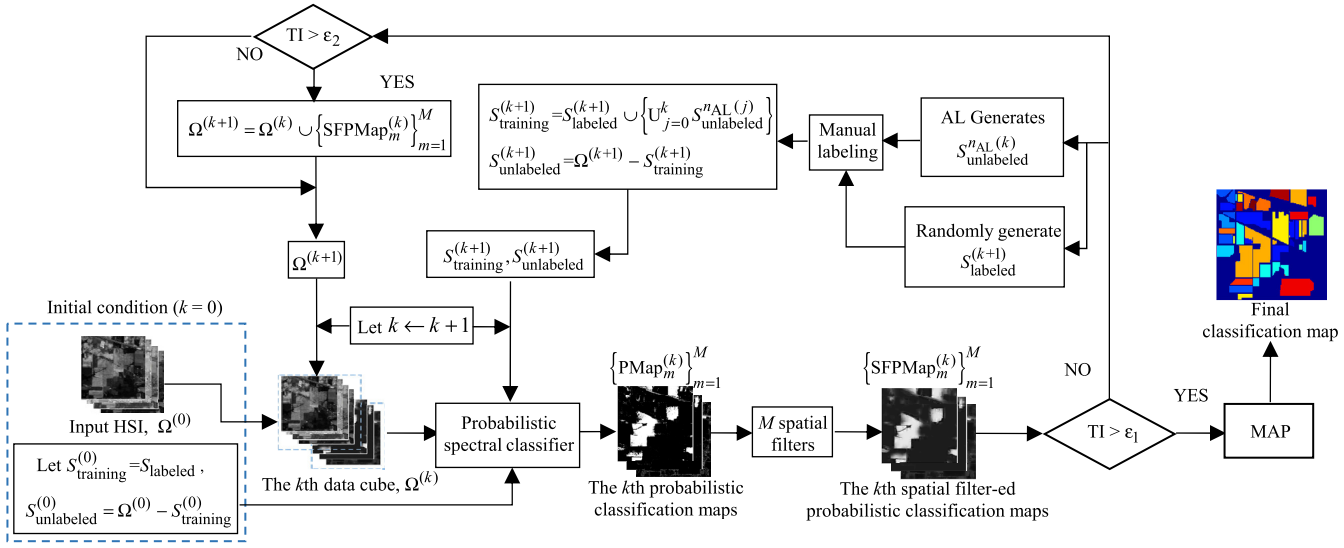


Fig. 2. Flowchart of IRTSA-AL-SS.

via MAP estimation. However, the optimization problem of solving $\{\mathbf{w}_m^{\mathrm{MAP}}\}_{m=1}^M$ is very complicated due to its nonquadratic and nonsmooth nature [16]. To address this problem, we take advantage of the LORSAL algorithm [16] which can be used to replace the difficult convex problem of solving $\{\mathbf{w}_m^{\mathrm{MAP}}\}_{m=1}^M$ with an easier quadratic optimization problem which significantly reduces the complexity of computing $\{\mathbf{w}_m^{\mathrm{MAP}}\}_{m=1}^M$. A detail implementation of MLR using LORSAL to learn $\{\mathbf{w}_m^{\mathrm{LORSAL}}\}_{m=1}^M$ can be found in [16]. Fig. 3 also depicts a diagram of finding MLR where $\mathbf{W} = [\mathbf{w}_1\mathbf{w}_2\cdots\mathbf{w}_M]$ and $\hat{\mathbf{W}} = [\mathbf{w}_1^{\mathrm{LORSAL}}\mathbf{w}_2^{\mathrm{LORSAL}}\mathbf{w}_M^{\mathrm{LORSAL}}]$.

The final spectral-based probability maps produced by MLR can be expressed as

$$\left\{\mathrm{PMap}_m\right\}_{m=1}^M = \left\{\left\{p_{\mathrm{MLR}}\left(\delta^{\mathrm{MLR}}(\mathbf{r}_n) = m|\mathbf{r}_n, \mathbf{W}\right)\right\}_{n=1}^N\right\}_{m=1}^M. \tag{15}$$

2) Spatial Filtered A Posteriori Probability Produced by Guided Filters: The guided filters [31] were originally derived from a local linear model which aims to smooth image while preserving the edge structure by considering a guidance image. This guidance image can be the image itself or another image. In this article, we imply first the PC image of HSI as the guidance image which is the same as the work demonstrated in [8].

More specifically, suppose sample $\mathbf{r}_n$ locates at pixel index (x_n, y_n), referred as $\mathbf{r}(x_n, y_n)$. An output *posteriori* spatial filtered probability result of $\mathbf{r}(x_n, y_n)$ using guided filter on MLR-ed maps can be represented as

$$p_{\mathrm{guided}}\left(\delta^{\mathrm{MLR}}(\mathbf{r}_n) = m|\mathbf{r}(x_n, y_n)\right) = \sum_{(x,y)} W_{(x,y)}(R)\, p_{\mathrm{MLR}}\left(\delta^{\mathrm{MLR}}(\mathbf{r}_n) = m|\mathbf{r}(x_n, y_n)\right). \tag{16}$$

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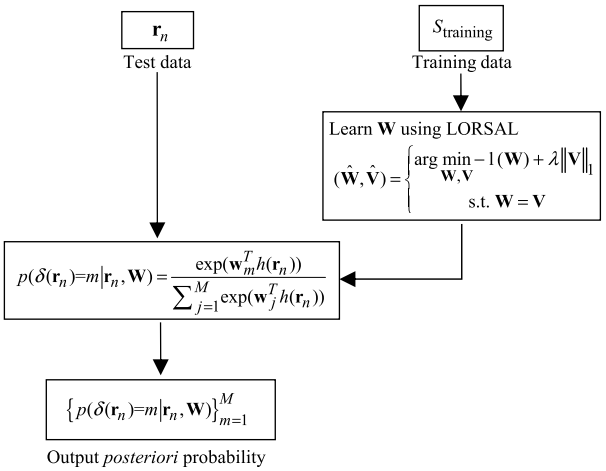


Fig. 3. Diagram of finding MLR.

The kernel function $W_{(x,y)}(R)$ in (13) only depends on the reference image R and is defined as

$$W_{(x,y)}(R) = \frac{1}{|\omega|}\sum_{k\in\omega_{(x,y)},k\in\omega_{(x_n,y_n)}}\left(1+\frac{(R(x,y)-\mu_k)(R(x_n,y_n)-\mu_k)}{\sigma_k^2+\varepsilon}\right) \tag{17}$$

where $\omega(x,y)$ and $\omega(x_n,y_n)$ indicate the local windows around pixel (x,y) and (x_n,y_n), respectively, $|\omega|$ is the size of filter window and the parameter ε is used to control the blurring effect of the guided filter. The output guided filtered probability maps is then obtained by

$$\left\{\text{SFPMap}_m\right\}_{m=1}^{M} = \left\{\left\{p_{\text{guided}}\left(\delta^{\text{MLR}}(\mathbf{r}_n)=m|\mathbf{r}_n\right)\right\}_{n=1}^{N}\right\}_{m=1}^{M} \tag{18}$$

and the binary classification maps using MLR with guided filters can be generated via MAP binarization in (11).

IV. Classification Measures

This section summarizes the classification measures [33], [34] to be used for experiments as follows.

1) M = number of classes.
2) n_{im} = number of samples in the mth class to be classified into the ith class.
3) n_{mm} = number of samples in the mth class correctly classified into the mth class.
4) n_m = number of data samples in the mth class, that is, $n_m=\sum_{i=1}^{M} n_{im}$.
5) $\hat{n}_{im}$ = number of data samples classified in the ith class, which are supposed to be in the mth class, C_j.
6) $\hat{n}_{ij}=n_{ij}$ with n_{ij} interpreted from *a priori* point of view and $\hat{n}_{ij}$ interpreted from *a posteriori* point of view.
7) $\hat{n}_m$ = number of data samples classified in the mth class, that is, $\hat{n}_m=\sum_{j=1}^{M}\hat{n}_{mj}$.
8) $\tilde{n}_m=\sum_{i=1,i\neq m}^{M} n_{im}$ = number of samples in the mth class with $i\neq m$ falsely classified into the mth class.
9) N = total number of data samples, $N=\sum_{m=1}^{M} n_m$.
10) C_m = set of samples in the mth class.
11) $\tilde{C}_m$ = set of data samples from classes $\{C_i\}_{i=1,i\neq m}^{M}$ classified into C_m.
12) $p(C_m)=(n_m/N)$.

In the following, we define several classification measures.

1) *Class Accuracy*

$$\begin{aligned}P_{\text{A}}(C_m) &= \text{accuracy of the } m \text{ th class}\\ &= P(C_{mm}|C_m)=\frac{n_{mm}}{n_m}.\end{aligned} \tag{19}$$

2) *Overall Precision Rate (PR)*

$$\begin{aligned}P_{\text{OPR}}\left(\hat{C}_m\right) &= p\left(\hat{C}_m|\text{classification}\right)\\ &= \frac{\hat{n}_{mm}}{\sum_{j=1}^{M}\hat{n}_{mj}}\end{aligned} \tag{20}$$

$$\begin{aligned}P_{\text{OPR}} &= \sum_{m=1}^{M} p\left(\hat{C}_m\right)p_{\text{PR}}\left(\hat{C}_m\right)\\ &= \frac{1}{N}\sum_{m=1}^{M}\hat{n}_m\, p_{\text{PR}}\left(\hat{C}_m\right).\end{aligned} \tag{21}$$

3) *OA*

$$\begin{aligned}P_{\text{OA}} &= p_{\text{overall-accuracy}}\left(\{C_m\}_{m=1}^{M}\right)\\ &= \sum_{m=1}^{M} p(C_m)p(C_{mm}|C_m)=\frac{1}{N}\sum_{m=1}^{M} n_m\left(\frac{n_{mm}}{\sum_{i=1}^{M} n_{im}}\right).\end{aligned} \tag{22}$$

4) *AA*

$$P_{\text{AA}}=\frac{1}{M}\sum_{m=1}^{M}P_{\text{A}}(C_m)=\frac{1}{M}\sum_{m=1}^{M}\frac{n_{mm}}{n_m}. \tag{23}$$

V. Real Images To Be Studied

Three well-known hyperspectral images, Purdue Indiana Indian Pines, Salinas, and University of Pavia, were used to demonstrate the advantage of using IFTSA-AL-SS and IRTSA-AL-SS. These images are available to be downloaded at http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.

A. Purdue Indiana Indian Pines

The first image is the Purdue Indiana India Pines as shown in Fig. 4(a)–(c). It is a well-known Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) image scene with a spatial resolution of 20 m per pixel and spectral resolution of 10 nm. It has a size of 145 × 145 pixel vectors and was recorded in June 1992 with 220 bands including water absorption bands (bands 104–108 and 150–162). In this article, the water absorption bands have been removed and only left with 200 bands for experiments. Table I tabulates all the 16 classes of interest along with the number of data samples in each class.

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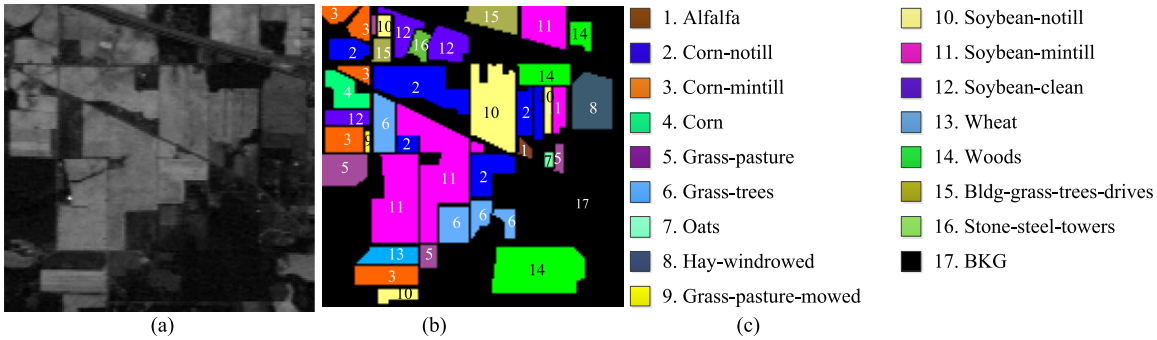


Fig. 4. AVIRIS image scene: Purdue Indiana Indian Pines test site. (a) Band 186 (2162.56 nm). (b) GT map. (c) Classes by colors.

TABLE I

CLASS LABELS OF PURDUE INDIANA INDIAN PINES WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (46)	Alfalfa	class 7 (28)	grass/pasture-mowed	class 13 (205)	wheat
class 2 (1428)	corn-notill	class 8 (478)	hay-windrowed	class 14 (1265)	woods
class 3 (830)	corn-min	class 9 (20)	Oats	class 15 (386)	bldg-grass green-drives
class 4 (237)	corn	class 10 (972)	soybeans-notill	class 16 (93)	stone-steel towers
class 5 (483)	grass/pasture	class 11 (2455)	soybeans-min		
class 6 (730)	grass/trees	class 12 (593)	soybeans-clean		

TABLE II

CLASS LABELS OF SALINAS SCENE WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (2009)	Brocoli_green_weeds_1	class 7 (3579)	Celery	class 13 (916)	Lettuce_romaine_6wk
class 2 (3726)	Brocoli_green_weeds_2	class 8 (11271)	Grapes_untrained	class 14 (1070)	Lettuce_romaine_7w
class 3 (1976)	Fallow	class 9 (6203)	Soil_vinyard_develop	class 15 (7268)	Vinyard_untrained
class 4 (1394)	Fallow_rough_plow	class 10 (3278)	Corn_senesced_green_weeds	class 16 (1807)	Vinyard_vertical_trellis
class 5 (2678)	Fallow_smooth	class 11 (1068)	Lettuce_romaine_4wk		
class 6 (3959)	Stubble	class 12 (1927)	Lettuce_romaine_5wk		

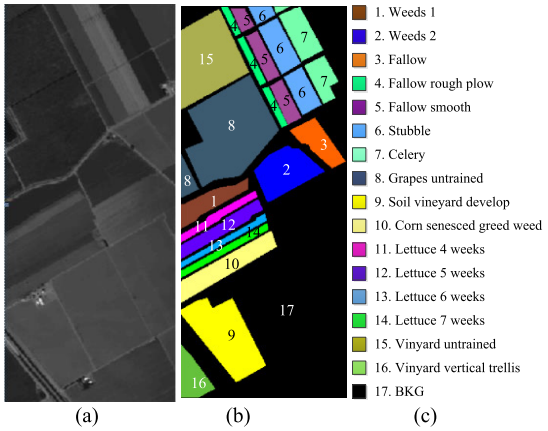


Fig. 5. GT of Salinas scene with 16 classes. (a) Salina scene. (b) GT map. (c) Class by colors.

B. Salinas

A second AVIRIS data used for experiments were Salinas scene as shown in Fig. 5(a), which was captured by the AVIRIS sensor over Salinas Valley, CA, USA, with a spatial resolution of 3.7 m per pixel in the wavelength range of 0.4–2.5 μm with a spectral resolution of 10 nm. It has a size of 512 $\times$ 217 $\times$ 224 (water absorption bands included). Fig. 5 shows the scene of the Salinas image along with the corresponding GT map in Fig. 5(b) along with 16 class labels plus background shown in Fig. 5(c). Same as the Purdue Indian Pines data, the water absorption bands have been removed, so only 204 spectral bands were left for conducting experiments. Table II tabulates the number of data samples in parentheses collected for each class.

C. University of Pavia

A third HSI data used for experiments were the University of Pavia image as shown in Fig. 6 which is an urban area surrounding the University of Pavia, Italy. It was recorded by the ROSIS-03 satellite sensor. It is of size 610 $\times$ 340 $\times$ 103 with a spatial resolution of 1.3 m per pixel and a spectral coverage ranging from 0.43 to 0.86 μm with a spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Nine classes of interest are considered for this image. Table III also tabulates the class labels and the number of data samples in each class of interest.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, three widely used hyperspectral image data sets discussed in Section V were used for experiments to evaluate the performance and effectiveness of IRTSA-AL-SS and IFTSA-AL-SS.

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TABLE III
CLASS LABELS OF UNIVERSITY OF PAVIA SCENE WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (6631)	Asphalt	class 4 (3064)	Trees	class 7 (1330)	Bitumen
class 2 (18649)	Meadows	class 5 (1345)	Painted metal sheets	class 8 (3682)	Self-Blocking Bricks
class 3 (2099)	Gravel	class 6 (5029)	Bare Soil	class 9 (947)	Shadows

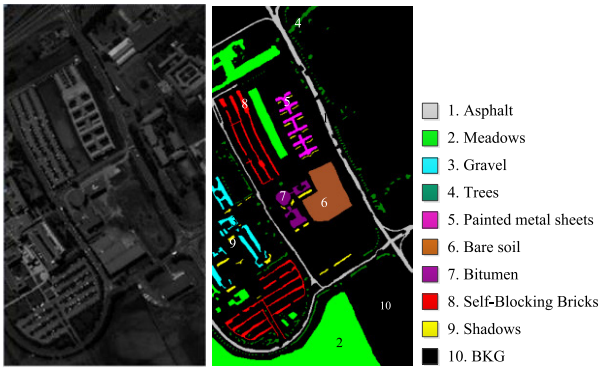


Fig. 6. GT of University of Pavia scene with nine classes. (a) University of Pavia scene. (b) GT. (c) Classes by colors.

As discussed in Section III, MLR was specifically chosen to be a spectral probabilistic classifier along with guided filters as SFs for comparison due to the fact that it has been widely used for AL [15]–[18]. In this case, IRTSA-AL-SS and IFTSA-AL-SS are particularly referred to as IRTSA-AL-MLR-Guided and IFTSA-AL-MLR-Guided which can be considered as generalized versions of MLR-Guided-AL, ITS-MLR-Guided, and IRTS-MLR-Guided where IRTS-MLR-Guided and ITS-MLR-Guided are the modified versions of IRTS-EPF-G-g [28] and IEPF-G-g [26] in the sense that the SVM used in IEPF is replaced by MLR.

To verify the effectiveness of IRTSA-AL-SS and IFTSA-AL-SS, the following experiments are designed as follows.

1) IFTSA-AL-MLR-Guided can be considered as an AL-augmented training sample version of the ITS-MLR-Guided and an ITS-expanded data cube version of the MLR-Guided-AL in which case these two methods were included for comparison. In addition, both ITS-MLR-Guided and IEPF-G-g are two supervised classifiers with the size of training samples in each class needed to be predetermined. In the following experiments, the sampling method used in [8] was also used to allocate the size of training samples for each class.
2) Moreover, three classic AL-SS approaches are also included and compared with IRTS-AL and IFTSA-AL. The first one is MLR-AL which simply uses MLR to produce *a posteriori* probability maps and then uses AL to augment training samples. A second one is MLR-MLL-AL [16] which uses MLR to produce initial probabilistic results and augment training samples via AL and then MLL with Markov random field MRF) to produce final classification maps. A third one is the MPM-LBP-AL [17] which implements LBP to refine the marginal probability maps by the AL-augmented training samples, and then the final classification results are produced by MAP marginal probability (MPM).
3) To reduce the uncertainty caused by RS of the training samples, the same experiment was conducted 15 times and the average along with the SD of the classification accuracy were calculated to evaluate the stability of all test algorithms.
4) In addition to the BT criterion, three other AL strategies, RS, MI [35], and MBT, were also used as alternative criteria to select informative unlabeled samples. Among them, RS randomly selected samples directly from an unlabeled sample set without using any criterion. The MI criterion developed in [16] and [35] was used to find unlabeled samples that maximize MI between a regressor and its class labels. The MBT [16] was used to increase the diversity of the BT-selected unlabeled training samples.
5) Most importantly, IRTSA/IFTSA-AL-SS can be implemented with any arbitrary classifier. Therefore, except MLR introduced in Section III, we also included experiments performed by IRTSA/IFTSA-AL-SS using three alternative spectral classifiers which are SVM [36], [37], extreme learning machine (ELM) 38], [39], and K-nearest neighbors (KNNs).

TABLE IV
PARAMETERS USED FOR AL-SS FOR THREE HSI DATA SETS

	n_{labeled}	n_{AL}	ε_1	ε_2	$k_{\max}$
Indian Pines	80	10	0.9985	0.93	32
Salinas	80	10	0.9985	0.85	22
U. Pavia	90	10	0.9985	0.85	21

A. Experimental Design Setting

The details of the parameters' setting used for the experiments are tabulated in Table IV where ε_1 and ε_2 are the two threshold values of TI used to terminate and start the feedback process where n_{labeled} is the size of initial labeled samples, n_{AL} is the number of unlabeled samples augmented by AL in each iteration, and $n^{\text{total}}_{\text{unlabeled}}$ is the total number of unlabeled samples included into S_{training} through entire iterative process.

To validate the effectiveness of IRTSA/IFTSA-AL, $n^{\text{total}}_{\text{unlabeled}}$ is a crucial criterion, where a better AL method can use a smaller $n^{\text{total}}_{\text{unlabeled}}$ to achieve higher classification accuracy.

One drawback that the traditional AL-SS algorithms have is no automatic stopping rule is provided to determine when the training sample augmentation process should be terminated. It is generally determined empirically by setting a maximum

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TABLE V

P_A, P_{OA}, P_{AA}, AND P_{OPR} OF SIX AL-SS ALGORITHMS USING BT STRATEGY WITH 80 INITIAL LABELED SAMPLES FOR PURDUE INDIAN PINES WITH HIGHEST ACCURACY BOLDFACED

Method	MLR-AL		MLR-MLL-AL		MPM-LPB-AL		MLR-Guided-AL		IFTSA-AL-MLR-Guided		IRTSA-AL-MLR-Guided	
$n_{\text{unlabeled}}^{\text{total}}$	320		320		320		320		310		**290**	
Class	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	95.12	35.78	95.12	**69.64**	97.44	37.62	97.56	48.78	**100**	16.88	**100**	17.02
2	75.69	61.84	83.36	**70.67**	98.52	69.02	98.98	66.75	99.12	63.13	**99.26**	66.98
3	76.52	40.29	96	56.15	92.13	67.04	97.96	70.38	**100**	70.6	99.88	**71.85**
4	63.39	29.83	98.66	52.74	97.73	59.07	**100**	56.46	**100**	**66.07**	**100**	65.88
5	86.81	30.38	85.11	31.82	98.72	25.11	**100**	24.68	**100**	23.59	**100**	**34.96**
6	97.49	33.61	**100**	39.37	**100**	47.35	97.08	**56.74**	**100**	47.38	**100**	44.25
7	95.65	31.43	95.65	**66.67**	**100**	43.14	**100**	30.99	95.65	59.46	**100**	40
8	98.93	74.68	**100**	71.23	**100**	**77.38**	**100**	72.73	**100**	71.3	**100**	72.29
9	**100**	6.42	78.57	**68.75**	**100**	26.32	**100**	23.81	**100**	32.61	**100**	30
10	74.59	54.01	79.91	**75.95**	99.46	66.21	98.69	59.78	**100**	61.36	99.78	61.78
11	81.49	44.56	95.89	47.27	98.9	65.52	96.1	**76.08**	99.92	74.96	**100**	69.72
12	80.68	64.52	96.96	**77.21**	98.75	64.19	98.21	65.75	**99.82**	57.38	99.65	59.48
13	99.5	63.99	99.5	**88.84**	99.5	71.07	99.5	70.57	**99.5**	74.44	**99.5**	70.97
14	95.2	32.43	96.56	33.08	99.84	32.71	99.19	**35.13**	**100**	28.14	99.92	26.83
15	65.58	9.49	79.13	8.75	75.69	8.05	100	8.97	**100**	15.47	**100**	**15.61**
16	73.26	45.65	81.4	46.98	82.56	**51.45**	91.67	20.7	**97.62**	32.03	97.56	17.02
P_{OA}/ P_{AA}	82.99/84.99		92.14/91.36		97.58/96.20		98.12/98.43		**99.81**/99.48		**99.81**/**99.72**	
P_{OPR}	38.88		43.16		45.71		45.96		46.80		**46.90**	

number of iterations, donated by $k_{\max}$, to end its iterative process. Therefore, $n_{\text{unlabeled}}^{\text{total}}$ in a classic AL-SS algorithm is predetermined with $n_{\text{unlabeled}}^{\text{total}} = k_{\max} \times n_{\text{AL}}$. In contrast, IRTSA/IFTSA-AL-SS determines $n_{\text{unlabeled}}^{\text{total}}$ by ε_1.

As shown in Table IV, n_{labeled} was set to 80 for Indian Pines and Salinas with five training samples for each class. On the other hand, since University of Pavia contains only nine classes, n_{labeled} was set to 90 with ten labeled samples for each class. Moreover, the $k_{\max}$ values for each image scene were 32 for the Indian Pines data and 22 for Salinas and 21 for University of Pavia. Accordingly, at the end, 400 training samples ($n_{\text{labeled}} = 80$ and $n_{\text{unlabeled}}^{\text{total}} = 320$) were used to classify Purdue Indian Pines and 300 training samples for both Salinas ($n_{\text{labeled}} = 80$ and $n_{\text{unlabeled}}^{\text{total}} = 220$) and University of Pavia ($n_{\text{labeled}} = 90$ and $n_{\text{unlabeled}}^{\text{total}} = 210$). In addition, the threshold values of ε_1 and ε_2 were selected empirically to achieve the best possible classification performance. Particularly, ε_2 plays an important role in controlling the effect of the used feedback process.

Finally, it should be noted that the number of training samples that is used to train the two supervised classifiers, ITS-MLR-Guided and IEPF-G-g, is identical to the final training samples number that AL-SS used, which was 400 for Indian Pines and 300 for both Salinas and U. Pavia. In this case, the performance of supervised and semisupervised algorithms could be fairly compared.

B. Comparative Analysis Among Test Methods

In the first set of experiments, a comparative analysis was conducted on six semisupervised methods and two supervised methods. To further evaluate the stability of the proposed algorithms, we run each of the same experiment 15 times to compute the average and SD in terms of their classification accuracies.

1) Purdue Indiana Indian Pines: To discuss the experimental results using IRTSA/IFTSA-AL, we first compare the classification performance among six semisupervised methods including IRTSA/IFTSA-AL-SS and four classic AL-SS algorithms. All the AL methods used BT as the training sample augmentation strategy with 80 initial labeled samples randomly selected from the GT. Table V tabulates $P_A(C_m)$, P_{OA}, P_{AA}, P_{OPR}, and $n_{\text{unlabeled}}^{\text{total}}$ resulting from single-run experiment of all test AL-SS methods.

According to Table V, the best classification result was produced by IRTSA-AL-MLR-Guided with $P_{OA}/P_{AA} = 99.81\%/99.72\%$ which was slightly better than IFTSA-AL-MLR-Guided with $P_{OA}/P_{AA} = 99.81\%/99.48\%$. However, IFTSA-AL-MLR-Guided required more augmented unlabeled samples ($n_{\text{unlabeled}}^{\text{total}} = 310$) to achieve the stopping rule compared with IRTSA-AL-MLR-Guided ($n_{\text{unlabeled}}^{\text{total}} = 290$). Specifically, the results in Table V also showed that IRTSA-AL-SS could effectively improve IFTSA-AL-SS by including IRTS to randomly reselect new $S_{\text{labeled}}^{(k)}$ in each iteration and by incorporating more *a posteriori* spatial information created by different sets of $S_{\text{labeled}}^{(k)}$ into data cube. This was mainly due to the fact that the classification inconstancy caused by different augmented training samples could be well-addressed via IRTSA-AL-SS, particularly when IFTSA-AL-SS is already shown to be very effective and improving its performance is considered to very difficult and challenging.

In addition, it clearly showed that all four classic AL-SS methods performed worse than IFTSA-AL-SS. This was because these AL-SS methods only augmented training samples without using feedback process to capture *posteriori* spatial information provided by spatial filtered probability maps. As observed from Table V, the worst AL-SS algorithm came from MLR-AL with $P_{OA}/P_{AA} = 82.99\%/84.99\%$ since it only used spectral information alone, and the three AL-SS classifiers all performed better than MLR-AL. Among them, MLR-MLL-AL was the worst one and MPM-LBP-AL performed better than it. However, the classification performance of MPM-LBP-AL was still worse than MLR-Guided-AL. This was because MPM-LBP used a small number of training

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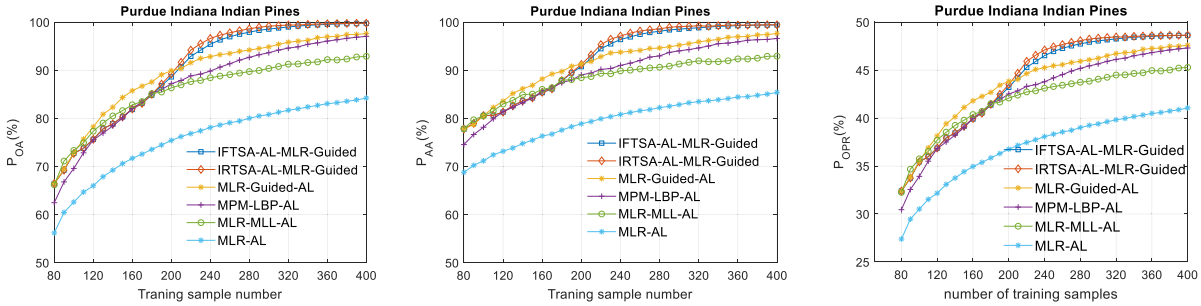


Fig. 7. P_{OA}, P_{AA}, and P_{OPR} using BT versus the number of training samples for Purdue Indiana Indian Pines data.

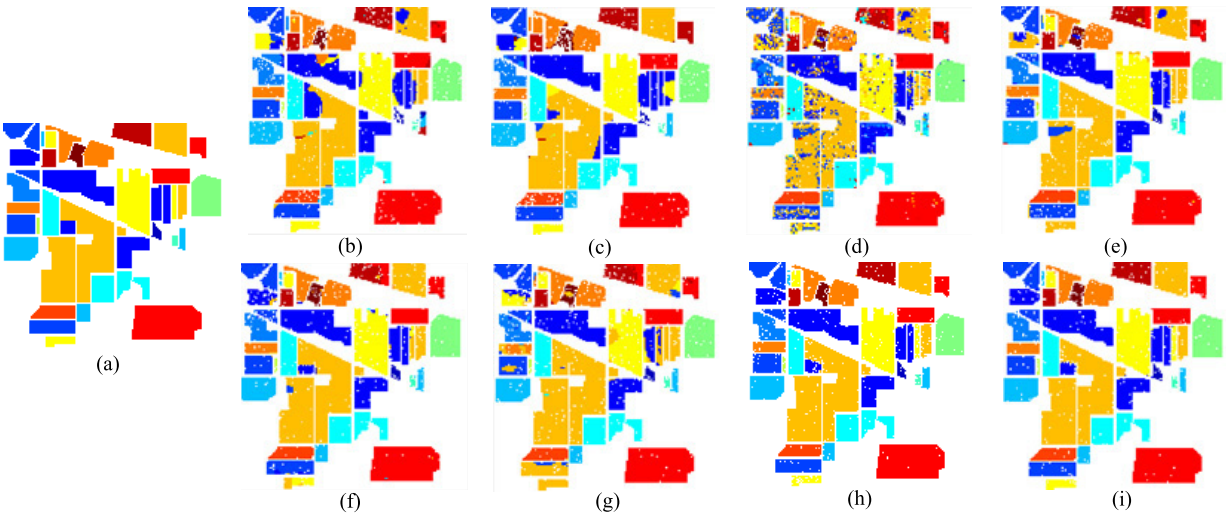


Fig. 8. Final classification maps for Indian Pines produced by eight test methods. (a) GT. (b) IEPF-G-g. (c) ITS-MLR-Guided. (d) MLR-AL. (e) MLR-MLL-AL. (f) MPM-LBP-AL. (g) MLR-Guided-AL. (h) IFTSA-MLR-Guided-AL. (i) IRTSA-AL-MLR-Guided.

sample labels that could not achieve good performance [17]. On the other hand, MLR-Guided-AL could achieve better performance by taking advantage of the *a posteriori* spatial information captured by guided filters after MLR classification even when used with small sizes of training samples. Nevertheless, all the AL-SS methods still performed worse than the IRTSA/IFTSA-AL methods. All the results in Table V demonstrated that IRTSA/IFTSA-AL-SS, which incorporated IRTS/IFTS to expand data cubes and AL to augment training sample sets, could indeed improve the classification performance of the traditional AL-SS significantly.

Fig. 7 plots the averaged P_{OA}, P_{AA}, and P_{OPR} resulting from 15-run experiments versus the use of different numbers of training samples for six AL-SS methods by starting with 80 initial labeled samples at the beginning and then using BT as an augmentation criterion to augment ten unlabeled training samples later at each iteration. Apparently, IRTSA/IFTSA-AL-MLR-Guided outperformed other four AL-SS algorithms using a smaller number of training samples to achieve better classification accuracies.

As shown in Fig. 7, when the number of training samples was smaller than 190, the P_{OA} curves of the two IRTSA/IFTSA-AL algorithms were similar to MLR-MLL and MPM-LBP-AL. However, when the number of training samples was greater than 190, P_{OA} of the IRTSA/IFTSA-AL-MLR-Guided improved significantly. This is because algorithms met the desired ε_2 and began its feedback process to augment filtered probability maps into the data cube. This indicated that the feedback processes implemented in IRTSA/IFTSA-AL played a crucial role in improving P_{OA}, P_{AA}, and P_{OPR}. In particular, IRTSA-AL-MLR-Guided performed slightly better than IFTSA-AL-MLR-Guided where a small gap between their P_{OA} curves at the range of n_{training} between 200 and 300 was visible. Fig. 8 and Table VI show the classification results of eight test methods including both supervised and semisupervised SS methods. By visually inspecting Fig. 8, it can be observed that IRTSA/IFTSA-AL-MLR-Guided produced more similar classification maps to the GT map of the Indian Pines scene as shown in Fig. 8(a). Table VI tabulates the averaged $P_{\mathrm{A}}(C_m)$, P_{OA}, and P_{AA} and their corresponding SD by implementing eight test HSIC methods 15 times where two ITS-SS-based methods, ITS-MLR-Guided and IEPF-G-g without using AL, obviously performed worse than MLR-Guided-AL and MPM-LBP-AL when the same set of training samples was used. This indicated that when the number of available training samples is small, adding informative unlabeled samples into a training sample set was more crucial than expanding the data cubes while keeping the same size of training samples. Nevertheless, IRTSA/IFTSA-AL combined both AL to augment training

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TABLE VI

AVERAGED P_A, P_{OA}, AND P_{AA} ALONG WITH SD OF EIGHT TEST ALGORITHMS FOR PURDUE INDIAN PINES WITH HIGHEST ACCURACY BOLDFACED

method	Supervised-HSIC		AL-SS				IFTSA-AL-SS	
	IEPF-G-g	ITS-MLR-Guided	MLR-AL	MLR-MLL-AL	MPM-LBP-AL	MLR-Guided-AL	IFTSA-AL-MLR-Guided	IRTSA-AL-MLR-Guided
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	98.26 (2.38)	**100** (0)	89.24 (5.46)	95.38 (1.55)	97.50 (1.73)	99.09 (1.26)	**100** (0)	**100** (0)
2	85.32 (5.95)	89.93 (4.42)	79.94 (3.54)	91.07 (5.50)	97.36 (2.22)	96.61 (2.60)	99.21 (0.31)	**99.33** (0.2)
3	90.28 (4.39)	96.23 (2.23)	70.40 (5.33)	80.37 (14.19)	91.74 (10.8)	91.6 (11.76)	**99.84** (0.13)	99.83 (0.22)
4	99.72 (0.42)	**100** (0)	68.75 (7.56)	95.59 (6.61)	99.18 (0.93)	98.89 (2.28)	**100** (0)	**100** (0)
5	94.7 (4.48)	97.11 (3.14)	92.06 (2.7)	95.30 (4.62)	98.97 (1.23)	99.92 (0.11)	99.95 (0.1)	99.95 (0.1)
6	97.95 (1.96)	98.95 (1.05)	97.05 (0.67)	**100** (0)	**100** (0)	99.34 (0.96)	**100** (0)	**100** (0)
7	94.29 (3.19)	95.72 (3.91)	87.74 (7.62)	97.80 (2.35)	97.19 (2.57)	96.14 (1.56)	**98.89** (2.06)	98.91 (2.01)
8	**100** (0)	**100** (0)	97.28 (4.21)	98.88 (3.18)	**100** (0)	**100** (0)	**100** (0)	**100** (0)
9	**100** (0)	**100** (0)	96.61 (3.63)	95.66 (8.33)	97.33 (3.65)	99.17 (2.36)	**100** (0)	**100** (0)
10	91.66 (4.05)	91.65 (2.86)	75.5 (3.45)	86.18 (6.78)	93.46 (5.16)	93.85 (6.87)	99.62 (0.46)	**99.72** (0.26)
11	83.86 (3.1)	92.52 (4.51)	83.52 (2.91)	97.31 (1.24)	98.80 (0.92)	98.52 (1.13)	**99.96** (0.06)	99.93 (0.14)
12	95.02 (5.33)	98.9 (0.7)	81.30 (4.31)	97.39 (1.69)	98.79 (1.05)	98.83 (0.33)	**99.67** (0.17)	**99.78** (0.12)
13	99.22 (0.75)	99.33 (0.47)	99.25 (0.38)	99.31 (0.37)	99.90 (0.22)	99.69 (0.26)	**99.62** (0.45)	99.37 (0.45)
14	98.39 (1.29)	97.74 (2.02)	92.04 (4.01)	94.42 (4.01)	99.45 (0.26)	99.47 (0.38)	**99.99** (0.03)	**100** (0)
15	99.83 (0.37)	**100** (0)	68.25 (7.14)	87.21 (10.83)	75.96 (5.90)	91.50 (9.91)	**99.97** (0.10)	**99.97** (0.10)
16	97.33 (2.87)	97.65 (1.68)	68.9 (10.62)	79.19 (14.15)	89.48 (8.00)	94.76 (5.68)	**97.93** (0.54)	97.87 (0.58)
P_{OA}	91.41 (1.16)	95.03 (1.60)	83.57 (0.56)	93.31 (1.74)	96.86 (1.04)	97.33 (1.47)	99.78 (0.04)	**99.80** (0.08)
P_{AA}	95.37 (1.03)	97.23 (0.57)	84.14 (1.29)	93.19 (1.72)	95.94 (0.48)	97.33 (1.19)	**99.67** (0.13)	**99.67** (0.14)
Time (s)	27.41 (5.69)	**15.29** (3.26)	20.75 (1.65)	28.30 (3.12)	896.33(50.6)	38.93 (3.63)	41.90 (6.52)	40.18 (3.16)

TABLE VII

AVERAGE NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF ALL TEST ALGORITHMS FOR PURDUE INDIAN PINES

method	Supervised-HSIC		AL-SS						IFTSA-AL-SS			
	IEPF-G-g/ ITS-MLR-Guided		MLR-AL/ MLR-MLL-AL		MPM-LBP-AL		MLR-Guided-AL		IFTSA-AL-MLR-Guided		IRTSA-AL-MLR-Guided	
Class	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$
1	23	23	5.5	40.5	6.4	39.6	5.25	40.75	6	40	6.25	39.75
2	27	1401	60.88	1367.13	68	1360	61.75	1366.25	64.5	1363.5	60.63	1367.38
3	28	802	46.75	783.25	44.2	785.8	44.38	785.63	34.88	795.13	31.38	798.63
4	25	212	17.38	219.63	20	217	21.75	215.25	15.25	221.75	14.5	222.5
5	25	458	16.13	466.88	18.6	464.4	18.13	464.88	18.63	464.38	17	466
6	27	703	14.38	715.63	14.4	715.6	10.75	719.25	13.25	716.75	12.38	717.63
7	14	14	5.38	22.63	6.4	21.6	5.38	22.63	6	22	5.63	22.38
8	26	452	9.63	468.38	7	471	6.63	471.38	9	469	8.13	469.88
9	10	10	5.38	14.63	5	15	5	15	5.13	14.88	5.13	14.88
10	28	944	51.38	920.63	47.6	924.4	58.13	913.88	49.13	922.88	42.88	929.13
11	31	2424	90.63	2364.38	75.6	2379.4	80.75	2374.25	76.5	2378.5	72.5	2382.5
12	28	565	32.63	560.38	31.2	561.8	28.25	564.75	25.75	567.25	25.88	567.13
13	26	179	5.13	199.88	6.2	198.8	5.25	199.75	6.88	198.13	7	198
14	28	1237	13.88	1251.13	17	1248	19.25	1245.75	17.38	1247.63	13.5	1251.5
15	28	358	18.25	367.75	22.6	363.4	21.38	364.63	19.5	366.5	15.25	370.75
16	26	67	6.75	86.25	9.8	83.2	8	85	8.5	84.5	10.75	82.25
$n_{labeled}$	400		80		80		80		80		80	
$n^{total}_{unlabeled}$	NA		320		320		320		296.25		268.75	
n_{test}	9849		9849		9849		9849		9872.75		9900.25	

sample sets and feedback process to expand data cubes were indeed superior to other test methods not only with the highest averaged P_{OA} and P_{AA} but also with smaller SD. For example, the best classification performance was produced by IRTSA-AL-MLR-Guided with the highest averaged $P_{OA}/P_{AA} =$ 99.80%/99.67%. Table VI also tabulates the running time of all test methods, where the fastest AL method was MLR-AL since it only required spectral classification without using any SF or regularization. In contrast, MPM-LBP-AL was the slowest method because it used iterative processes to find the marginal probabilities of each data sample which was very time-consuming. Interestingly, although IRTSA/IFTSA-AL-SS needed additional feedback loops, it would not take longer computing time than traditional AL methods. As an example, the average running time required by IFTSA-AL-MLR-Guided was 41.9 s which was only 3 s longer than its counterpart, MLR-Guided-AL. This was a great advantage of IRTSA/IFTSA-AL-SS which could significantly improve the classification performance of AL-SS at the expense of slightly higher computing time.

Finally, Table VII tabulates the average number of training samples and test samples for each class after AL augmentation processes and $n^{total}_{unlabeled}$ of each test method. It is clear to see that IRTSA-AL-MLR-Guided showed its efficiency in selecting appropriate training samples that only required an average 268.75 $n^{total}_{unlabeled}$ to achieve the stopping rule, which terminated faster than IFTSA-AL-MLR-Guided that needed average 296.25 $n^{total}_{unlabeled}$ to terminate the feedback process.

2) Salinas: Fig. 9 shows the single-run classification maps along with their averaged P_A, P_{OA}, and P_{AA} values obtained by 15-run experiments tabulated in Table VIII for all test

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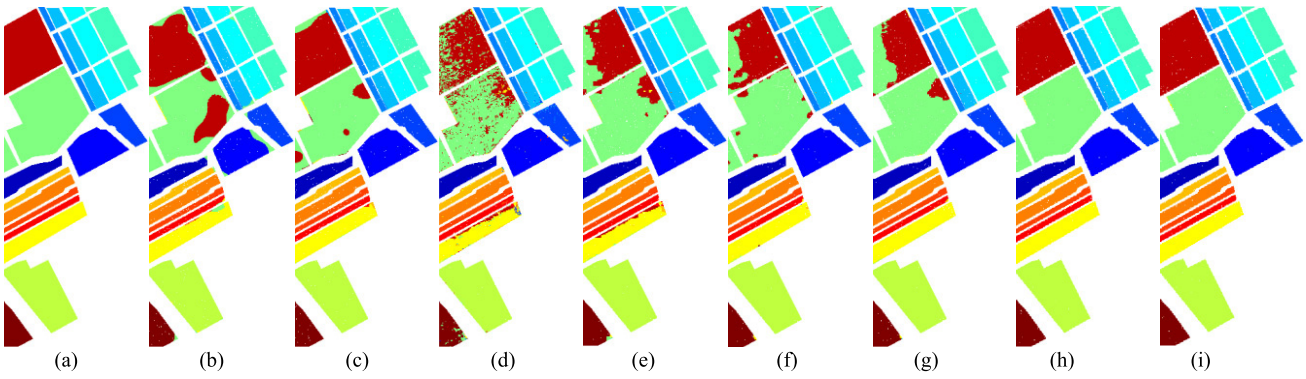


Fig. 9. Final classification maps for Salinas produced by eight test methods. (a) GT. (b) IEPF-G-g. (c) ITS-MLR-Guided. (d) MLR-AL. (e) MLR-MLL-AL. (f) MPM-LBP-AL. (g) MLR-Guided-AL. (h) IFTSA-MLR-Guided-AL. (i) IRTSA-AL-MLR-Guided.

TABLE VIII

AVERAGED P_A, P_{OA}, AND P_{AA} ALONG WITH SD OF EIGHT TESTED METHODS FOR SALINAS WITH HIGHEST ACCURACY BOLDFACED

method	Supervised-HSIC		AL-SS				IFTSA-AL-SS	
	IEPF-G-g	ITS-MLR-Guided	MLR-AL	MLR-MLL-AL	MPM-LBP-AL	MLR-Guided-AL	IFTSA-AL-MLR-Guided	IRTSA-AL-MLR-Guided
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	**100** (0)	**100** (0)	97.85 (2.09)	99.35 (1)	99.66 (0.47)	99.98 (0.07)	**100** (0)	**100** (0)
2	99.81 (0.31)	**100** (0)	99.43 (0.51)	99.82 (0.25)	99.79 (0.24)	**100** (0)	**100** (0)	**100** (0)
3	99.44 (0.78)	99.98 (0.04)	97.17 (2.07)	99.36 (1.41)	99.74 (0.14)	**100** (0)	99.97 (0.09)	**100** (0)
4	99.54 (0.29)	99.91 (0.12)	99.39 (0.49)	99.39 (0.49)	99.17 (0.54)	99.69 (0.49)	**99.98** (0.05)	99.97 (0.08)
5	98.43 (0.45)	98.81 (0.27)	97.05 (0.71)	98.71 (0.52)	99.17 (0.28)	98.88 (0.28)	99.13 (0.11)	**99.36** (0.13)
6	99.95 (0.05)	99.95 (0.03)	98.75 (2.25)	99.2 (1.77)	99.84 (0.07)	99.98 (0.03)	99.99 (0.02)	**100** (0.01)
7	99.64 (0.13)	99.92 (0.02)	99.49 (0.3)	99.71 (0.27)	99.66 (0.3)	99.93 (0.06)	99.96 (0.05)	**99.99** (0.02)
8	90.9 (4.18)	85.12 (7.92)	86.54 (1.76)	93.7 (2.57)	94.41 (2.58)	94.07 (3.64)	99.86 (0.2)	**99.95** (0.09)
9	96.81 (2.6)	99.91 (0.18)	99.66 (0.31)	99.95 (0.06)	99.99 (0.01)	99.98 (0.05)	**100** (0)	**100** (0.01)
10	96.13 (3.42)	96.1 (1.26)	91.13 (4.62)	93.89 (3.59)	98.72 (0.29)	97.38 (4.4)	**99.97** (0.05)	**99.97** (0.04)
11	98.68 (1.07)	99.83 (0.29)	95.75 (2.54)	97.27 (2.11)	99.77 (0.41)	99.55 (0.5)	**100** (0)	99.98 (0.07)
12	**100** (0)	**100** (0)	99.79 (0.18)	99.99 (0.02)	99.99 (0.02)	**100** (0)	**100** (0)	**100** (0)
13	99.02 (0.68)	99.26 (0.45)	98.3 (1.04)	98.45 (0.86)	98.7 (0.86)	99.18 (0.39)	99.41 (0.33)	**99.74** (0.26)
14	98.1 (1.14)	98.89 (0.54)	92.05 (4.73)	95.52 (4.1)	98.53 (1.32)	99.21 (0.57)	99.58 (0.58)	**99.61** (0.3)
15	88.78 (6.16)	96.18 (3.75)	65.17 (5.33)	71.59 (6.76)	78.96 (2.92)	94.5 (4.5)	99.71 (0.44)	**99.94** (0.07)
16	98.31 (0.44)	**100** (0)	95.72 (2.63)	98.8 (0.57)	96.89 (2.78)	99.28 (0.42)	**100** (0)	**100** (0)
P_{OA}	95.70 (1.33)	96.02 (1.79)	91.05 (0.46)	94.10 (0.59)	95.66 (0.23)	97.74 (0.62)	99.86 (0.07)	**99.93** (0.02)
P_{AA}	97.72 (0.52)	98.37 (0.62)	94.58 (0.41)	96.54 (0.30)	97.69 (0.22)	98.85 (0.37)	99.85 (0.05)	**99.91** (0.02)
Time (s)	62.8 (24.67)	48.35 (17.2)	37.70 (1.67)	91.32 (2.56)	3060.1 (79.5)	115.87 (7.48)	135.8 (24.36)	108.99 (24.46)

methods. Table IX also tabulates the averaged training sample numbers for each class that were augmented by AL. Once again, IRTSA/IFTSA-AL both produced closer maps to the GT map in Fig. 9(a) than those produced by other test methods where the worst classification result was produced by MLR-AL.

By examining Table VIII, the two best classifiers were IRTSA-AL-MLR-Guided and IFTSA-AL-MLR-Guided with averaged $P_{OA}/P_{AA} = 99.93\%/99.91\%$ and $99.86\%/99.85\%$ where IRTSA-AL performed slightly better than IFTSA-AL in terms of averaged P_{AA}. Specifically, there were obvious improvement in P_A in classification results of classes 8 and 15 where $P_A(C_8)$ produced by MPM-LBP-AL and MLR-Guided-AL were 94.41% and 94.07% which were improved nearly 5% to 99.86% and 99.95% by IFTSA-AL-MLR-Guided and IRTSA-AL-MLR-Guided. Moreover, although the performance of IRTSA-AL and IFTSA-AL were very close, from their $n^{\text{total}}_{\text{unlabeled}}$ tabulated in Table IX, IRTSA-AL-MLR-Guided only required averaged $n^{\text{total}}_{\text{unlabeled}} = 146.25$ to achieve the stopping criterion was which more effective than IFTSA-AL-MLR-Guided which needed $n^{\text{total}}_{\text{unlabeled}} = 155$ to terminate its iterative process. This indicated that IRTSA-AL was superior to IFTSA-AL in the sense that the former was able to use fewer training samples to achieve better classification accuracy that the latter.

Finally, Fig. 10 plots the averaged P_{OA}, P_{AA}, and P_{OPR} versus different numbers of training samples used for the Salinas data where AL used BT as a criterion for training sample augmentation and n_{labeled} was set to 80. Clearly, IRTSA/ITSA-MLR-Guided-AL produced higher accuracies than other test AL methods. Like the Indian Pines results, IRTSA/IFTSA-AL only showed slight improvement compared with other classic AL methods in the early iterative stages before algorithm actually started to feed back probability maps. Once IRTSA/IFTSA-AL began their feedback processes, their performance in classification started to improve significantly.

3) University of Pavia: Fig. 11 shows the final classification maps of University of Pavia with Table X which tabulates the averaged classification accuracies along with SD resulting from 15-run experiments. According to Fig. 10, the same conclusions drawn from previous experiments were also applied to this scene where the worst method was MLR-AL and

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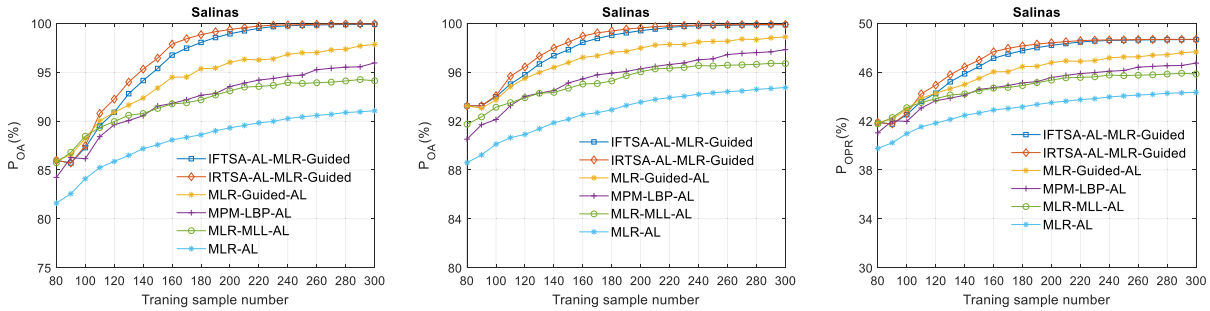


Fig. 10. P_{OA}, P_{AA}, and P_{OPR} of six AL-SS methods using BT versus different numbers of training samples for Salinas data set.

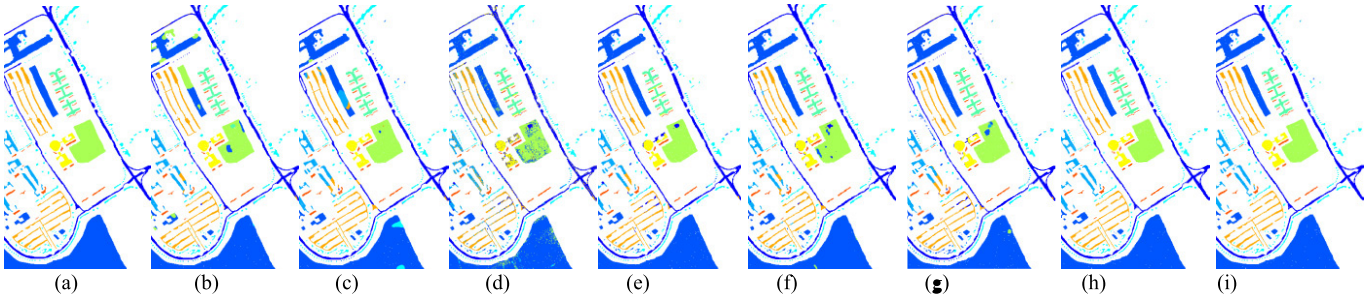


Fig. 11. Final classification maps for University of Pavia produced by eight test methods. (a) GT. (b) IEPF-G-g. (c) ITS-MLR-Guided. (d) MLR-AL. (e) MLR-MLL-AL. (f) MPM-LBP-AL. (g) MLR-Guided-AL. (h) ITSA-MLR-Guided-AL. (i) IRTSA-AL-MLR-Guided.

TABLE IX

AVERAGE NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF EIGHT TESTED ALGORITHMS FOR SALINAS

method	Supervised-HSIC		AL-SS						IFTSA-AL-SS			
	IEPF-G-g/ ITS-MLR-Guided		MLR-AL/ MLR-MLL-AL		MPM-LBP-AL		MLR-Guided-AL		IFTSA-AL-MLR-Guided		IRTSA-AL-MLR-Guided	
Class	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$	$n_{\mathrm{train}}(C_m)$	$n_{\mathrm{test}}(C_m)$
1	19	1990	6.25	2002.75	7	2002	5.13	2003.88	5.38	2003.63	5.38	2003.63
2	19	3707	7.13	3718.88	6.6	3719.4	5	3721	6.13	3719.88	6.5	3719.5
3	19	1957	14.25	1961.75	9.8	1966.2	7	1969	7.88	1968.13	8	1968
4	18	1376	6	1388	6.8	1387.2	5.25	1388.75	8.5	1385.5	11.13	1382.88
5	19	2659	10.88	2667.13	9	2669	9.75	2668.25	12.38	2665.63	14.38	2663.63
6	18	3941	6.63	3952.38	7.6	3951.4	5.75	3953.25	6.88	3952.13	7.63	3951.38
7	19	3560	6.13	3572.88	6	3573	5.63	3573.38	7.25	3571.75	7.38	3571.63
8	19	11252	94.38	11176.63	96.2	11174.8	102.13	11168.88	64.25	11206.75	61.75	11209.25
9	19	6184	9.25	6193.75	8.8	6194.2	5.88	6197.13	6.5	6196.5	7	6196
10	18	3260	18	3260	21.2	3256.8	16.5	3261.5	17	3261	16.38	3261.63
11	19	1049	9	1059	11	1057	5.38	1062.63	6.88	1061.13	7.13	1060.88
12	18	1909	6	1921	6	1921	5.5	1921.5	5.63	1921.38	5.75	1921.25
13	19	897	5.38	910.63	6.4	909.6	5.75	910.25	6.5	909.5	7	909
14	19	1051	8.13	1061.88	10.4	1059.6	8.5	1061.5	10.5	1059.5	10.5	1059.5
15	19	7249	83.5	7184.5	79.2	7188.8	98.63	7169.38	53.63	7214.38	41.38	7226.63
16	19	1788	9.13	1797.88	8	1799	8.25	1798.75	9.75	1797.25	9	1798
n_{labeled}	300		80		80		80		80		80	
$n_{\mathrm{unlabeled}}^{\mathrm{total}}$	NA		220		220		220		155		146.25	
n_{test}	53829		53829		53829		53829		53894		53902.75	

the best classification result was once again produced by IRTSA/IFTSA-MLR-Guided-AL. As shown in Table X, the highest accuracy was IRTSA-MLR-Guided-AL with averaged $P_{\mathrm{OA}}/P_{\mathrm{AA}} = 99.85\%/99.71\%$ followed by IFTSA-MLR-Guided-AL which was the second best with $P_{\mathrm{OA}}/P_{\mathrm{AA}} = 99.71\%/99.46\%$.

Table XI also tabulates the average number of training and test samples for each of nine classes where IFTSA-AL-MLR-Guided required $n_{\mathrm{unlabeled}}^{\mathrm{total}} = 205$ which was much higher than $n_{\mathrm{unlabeled}}^{\mathrm{total}} = 155$ required by IRTSA-AL-MLR-Guided before their algorithms were terminated. These results were also very similar to that obtained for Salinas and Indian Pines. That is, although both IRTSA/IFTSA-AL produced similar classification accuracies, IRTSA-AL was more efficient than IFTSA-AL-SS using fewer training samples to achieve better classification performance.

Finally, Fig. 12 plots P_{OA}, P_{AA}, and P_{OPR} versus different numbers of training samples for the University of Pavia data

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TABLE X

AVERAGED P_A, P_{OA}, AND P_{AA} ALONG WITH SD OF SIX AL-SS ALGORITHMS FOR UNIVERSITY OF PAVIA WITH HIGHEST ACCURACY BOLDFACED

method	Supervised-HSIC		AL-SS				IFTSA-AL-SS	
	IEPF-G-g	ITS-MLR-Guided	MLR-AL	MLR-MLL-AL	MPM-LBP-AL	MLR-Guided-AL	IFTSA-AL-MLR-Guided	IRTSA-AL-MLR-Guided
Class	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$	$P_A(C_m)$
1	95.71 (1.64)	97.96 (0.73)	84.78 (4.05)	97.51 (1.95)	96.58 (1.86)	97.97 (0.65)	99.87 (0.36)	**100** (0.01)
2	86.92 (6.41)	93.73 (3.86)	95.78 (1.95)	99.29 (1.2)	99.77 (0.09)	99.76 (0.23)	**99.99** (0.02)	**99.98** (0.02)
3	95.6 (1.2)	94.9 (2.5)	72.49 (7.7)	78.15 (13.03)	88.53 (4.79)	92.84 (3.46)	98.42 (2.85)	**99.79** (0.59)
4	94.34 (1.59)	94.74 (2.06)	86.28 (2.75)	88.47 (1.96)	96.59 (2.19)	94.33 (1)	98.39 (0.66)	**98.65** (0.25)
5	99.95 (0.1)	99.97 (0.07)	95.08 (4.01)	97.2 (3.03)	99.56 (0.36)	99.93 (0.1)	**100** (0)	**99.98** (0.05)
6	96.39 (5.05)	99.73 (0.42)	80.24 (3.55)	94.44 (5)	97.82 (2.77)	97.6 (2.42)	**100** (0)	**100** (0)
7	**100** (0)	**100** (0)	82.82 (5.66)	87.15 (11.53)	97.05 (0.89)	99.17 (0.96)	**100** (0)	**100** (0)
8	96.06 (2.95)	96.57 (2.24)	85.18 (2.89)	94.66 (4.32)	96.12 (2.76)	96.75 (2.74)	99.41 (0.67)	**99.75** (0.64)
9	99.96 (0.1)	**100** (0)	99 (0.91)	98.27 (2.18)	99.44 (0.47)	**99.93** (0.11)	99.04 (0.62)	99.19 (0.83)
P_{OA}	92.22 (2.92)	95.99 (1.67)	89.18 (1.05)	95.78 (1.00)	97.86 (0.35)	98.24 (0.50)	99.71 (0.20)	**99.85** (0.09)
P_{AA}	96.10 (0.83)	97.51 (0.33)	86.85 (0.86)	92.79 (2.22)	96.83 (0.32)	97.59 (0.72)	99.46 (0.37)	**99.71** (0.19)
Time (s)	41.15 (5.24)	47.08 (5.27)	49.65 (4.32)	111.48 (6.4)	5203.3 (13.5)	151.67 (7.3)	168.64 (36.6)	126.05 (16.1)

TABLE XI

AVERAGE NUMBER OF TRAINING SAMPLES AND TEST SAMPLES FOR EACH CLASS OF ALL TEST ALGORITHMS FOR UNIVERSITY OF PAVIA

method	Supervised-HSIC		AL-SS						IFTSA-AL-SS			
	IEPF-G-g/ ITS-MLR-Guided		MLR-AL/ MLR-MLL-AL		MPM-LBP-AL		MLR-Guided-AL		IFTSA-AL-MLR-Guided		IRTSA-AL-MLR-Guided	
Class	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$	$n_{train}(C_m)$	$n_{test}(C_m)$
1	33	6598	53.5	6577.5	45	6586	47	6584	44.13	6586.88	39.5	6591.5
2	34	18615	63.75	18585.25	53.8	18595.2	43.25	18605.75	59.25	18589.75	53.5	18595.5
3	34	2065	34.13	2064.88	37.8	2061.2	44.5	2054.5	41.25	2057.75	24	2075
4	33	3031	16.88	3047.13	28.8	3035.2	43.38	3020.63	45.63	3018.38	32.75	3031.25
5	33	1312	10.75	1334.25	12.8	1332.2	10.63	1334.38	12.25	1332.75	13.13	1331.88
6	33	4996	46.75	4982.25	36	4993	34.25	4994.75	24.88	5004.13	24.5	5004.5
7	33	1297	24.38	1305.63	26	1304	21.13	1308.88	13.75	1316.25	11.75	1318.25
8	34	3648	39.25	3642.75	48.2	3633.8	45.25	3636.75	41.13	3640.88	33.25	3648.75
9	33	914	10.63	936.38	11.6	935.4	10.63	936.38	12.75	934.25	12.63	934.38
$n_{labeled}$	90		90		90		90		90		90	
$n_{unlabeled}^{total}$	210		210		210		210		205		155	
n_{test}	42476		42476		42476		42476		42481		42531	

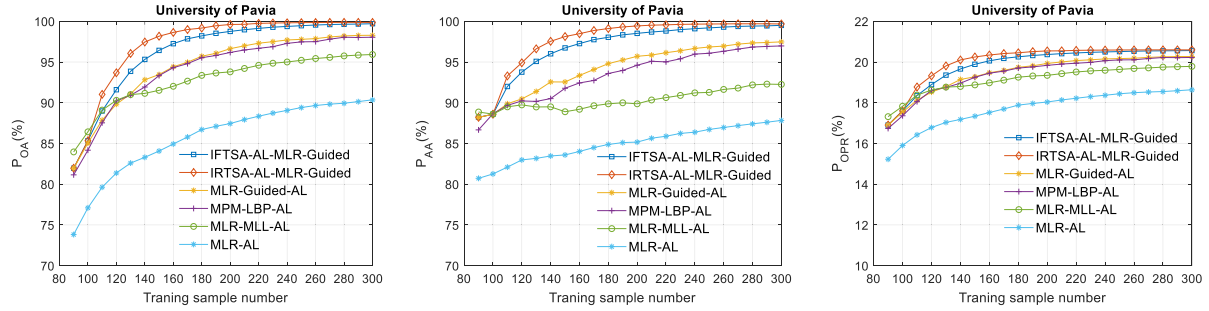


Fig. 12. P_{OA}, P_{AA}, and P_{OPR} of six AL-SS methods using BT versus different numbers of training samples for University of Pavia data set.

where AL used BT as a criterion and $n_{labeled}$ was set to 90. The conclusion was similar to that drawn for two previous image scenes where IRTSA/IFTSA-MLR-Guided-AL clearly produced higher accuracies than other test AL methods.

C. Comparative Analysis for Different Values of ε_2

One crucial issue of implementing ITSA-AL is how to select appropriate value of ε_2. To illustrate the influence of ε_2 in terms of final classification performance, Fig. 13 plots the P_{OA} curves of single-run IRTSA/IFTSA-AL using five different values of ε_2.

First, if we compare the P_{OA} curves resulting from IFTSA-AL using $\varepsilon_2 = 0.98$ and 0.93 for Indian Pines, it was clear that their final P_{OA} values were almost identical but $\varepsilon_2 = 0.98$ required more augmented training samples to terminate the iterative process. That is, when ε_2 was too large, the algorithm required lots of iterations to start its feedback process which in turn also required more iterations to terminate the algorithm.

On the other hand, when a small ε_2 (i.e., $\varepsilon_2 = 0.75$ or 0.83) value was used, P_{OA} increased faster at the very beginning of the algorithms but may also decrease their final P_{OA} when algorithms achieved the stopping condition. For example,

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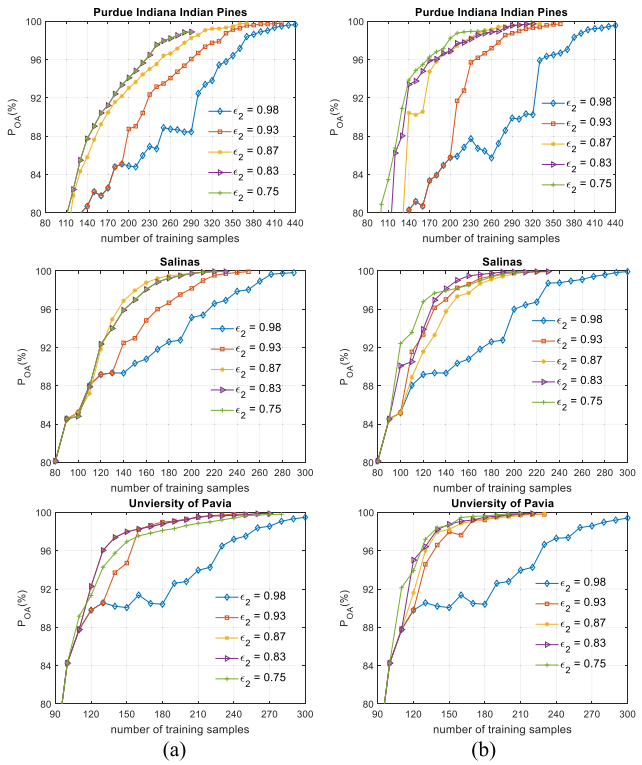


Fig. 13. P_{OA} versus the number of training samples using various ε_2 values. (a) IFTSA-AL-MLR-Guided. (b) IRTSA-AL-MLR-Guided.

the final P_{OA} of IFTSA-AL using $\varepsilon_2 = 0.75$ was obviously lower than using a larger ε_2 (i.e., $\varepsilon_2 = 0.87$, 0.93, and 0.98) value in Indian Pines experiments. This means that if the feedback process began too early, the augmented spatial filtered maps produced using a small number of training samples would also contain a high rate of misclassification errors, and thus decrease the final P_{OA}.

Overall, ε_2 could be a significant factor in affecting the final classification results. If ε_2 is too large, the algorithms would augment too many training samples to satisfy the stopping criterion. On the other hand, a small value of ε_2 would lead to a degradation of the final P_{OA}. In this article, the value of ε_2 selected for the three data sets was selected empirically as tabulated in Table IV. Nevertheless, according to our experience, an appropriate value of ε_2 falls in a range from 0.87 to 0.93 which would be a safe selection for the three data sets to be able to achieve high accuracy without using too many training samples. A future work will focus on how to automatically decide ε_2 for various data sets.

D. Comparative Analysis for Different AL Criteria

To investigate the advantages and disadvantages of AL methods over IRTSA/IFTSA-AL, four AL criteria, BT, MBT, MI, and RS, were investigated to be jointly implemented with IRTSA/IFTSA-AL. For fair comparison, the identical parameter setting tabulated in Table IV was used for all AL criteria. It should be noted that MBT selects its unlabeled training samples uniformly and augments each individual class with $n_{\mathrm{AL}} = c \times M$ where c is the number of training samples to be augmented into each class in one iteration. It is quite different from other AL criteria which set n_{AL} to any positive integer. So, in the following experiments, n_{AL} was set to M for MBT with $c = 1$ and $n_{\mathrm{AL}} = 10$ for other AL criteria.

Fig. 14 plots P_{OA} versus the use of different numbers of training samples using IFTSA-AL-MLR-Guided and IRTSA-AL-MLR-Guided in conjunction with different AL criteria for three image scenes.

The results of Indian Pines in Fig. 14 show that the best AL criterion was BT which always produced the highest P_{OA} among all AL criteria and MI was the second best because its improvement in P_{OA} with the increase in training samples was slower than BT. It was also worth noting that when MI was used, the selected training samples might trap in the most complex area (i.e., $\mathbf{p}(\delta(\mathbf{r}_n)|\mathbf{r}_n)$) is close to be uniform), as opposed to BT that has high diversity in selecting training samples. Moreover, MBT did not perform well on this image scene mainly due to the fact that MBT uniformly augmented training samples to each class, and thus, some imbalanced classes such as class 9 with only 20 data samples might have an oversampling problem. Therefore, the performance of MBT was even worse than RS until sufficient number of training samples was augmented into the training sample sets.

For Salinas and University of Pavia, once again the best results were produced by BT. On the other hand, even though MI and MBT did not performed well coupled with IFTSA-AL-MLR-Guided, their P_{OA} was significantly improved if IRTSA-AL-MLR-Guided was used. For instance, MI could achieve almost identical P_{OA} plot as BT for Salinas using IRTSA-AL-MLR-Guided compared with MI coupled with IFTSA-AL-MLR-Guided which produced lower P_{OA} than BT.

Table XII further tabulates the averaged P_{OA}, P_{AA}, and $n^{\mathrm{total}}_{\mathrm{unlabeled}}$ along with its SD of IRTSA/IFTSA-AL using four AL criteria where RS was the worst one which not only used the highest $n^{\mathrm{total}}_{\mathrm{unlabeled}}$ but also produced the lowest $P_{\mathrm{OA}}/P_{\mathrm{AA}}$.

For the experiments conducted for the Purdue Indian Pines and University of Pavia, BT always achieved the best classification performance with the highest $P_{\mathrm{OA}}/P_{\mathrm{AA}}$ and the lowest $n^{\mathrm{total}}_{\mathrm{unlabeled}}$. On the other hand, for the Salinas, the results showed that the performance of BT and MI was very close where BT used a smaller $n^{\mathrm{total}}_{\mathrm{unlabeled}}$ to terminate the iterative process but MI was able to achieve the slightly higher $P_{\mathrm{OA}}/P_{\mathrm{AA}}$ than BT. However, in terms of two P_{OA} curves plotted in Fig. 10, BT was still superior to MI since it had faster improvement in P_{OA} as the number of training samples was augmented by AL criteria. Overall, the experimental results clearly demonstrated the advantage of selecting BT as the desired AL criterion to be implemented in conjunction with IRTSA/IFTSA.

E. Comparative Analysis for Different Spectral-Classifiers

In general, IRTSA-AL and IFTSA-AL have a generic framework which can implement any arbitrary SS classifier coupled with AL not only to expand data cubes but also to augment training samples. As demonstrated in the experiments conducted in this article, three other classifiers, SVM, ELM, and KNN, were chosen to be implemented in conjunction with

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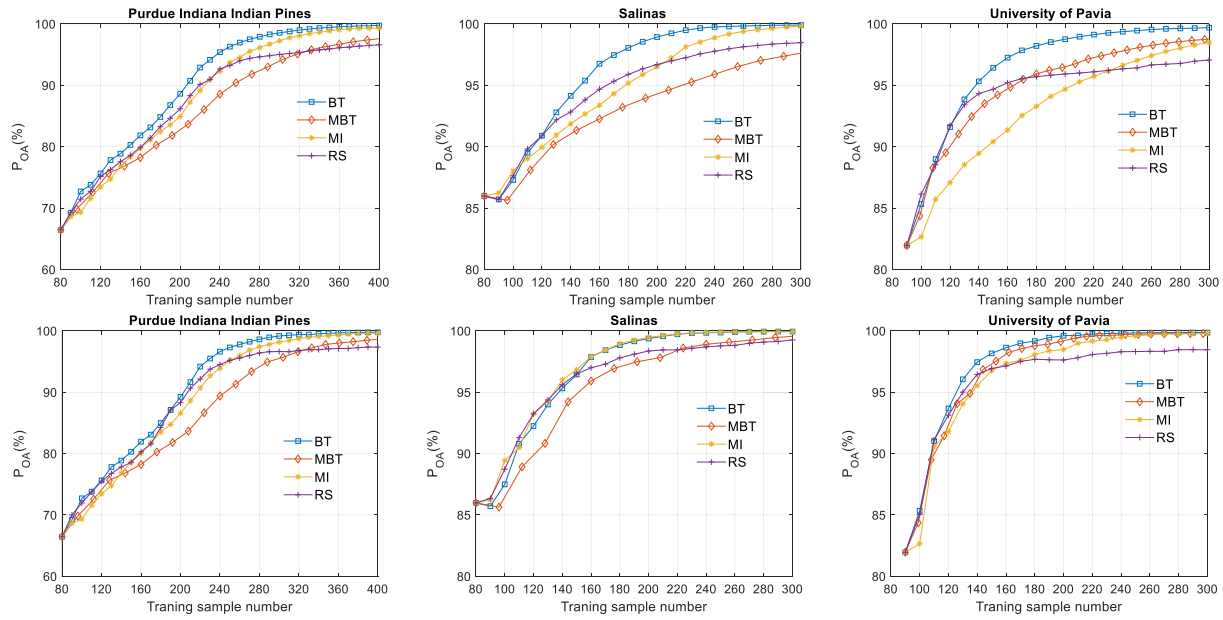


Fig. 14. P_{OA} of (First row) IFTSA-AL-MLR-Guided and (Second row) IRTSA-AL-MLR-Guided using four AL strategies versus training sample number for three HSI data sets.

TABLE XII

P_{A}, P_{OA}, P_{AA}, AND $n_{\mathrm{ULABELED}}^{\mathrm{TOTAL}}$ P_{OA}, P_{AA} AND OF TWO IFTSA-AL METHODS USING FOUR DIFFERENT AL STRATEGIES FOR THREE HSI WITH HIGHEST ACCURACY BOLDFACED

		IFTSA-AL-MLR-Guided				IRTSA-AL-MLR-Guided			
		BT	MBT	MI	RS	BT	MBT	MI	RS
Indian Pines	$n_{\mathrm{ulabeled}}^{\mathrm{total}}$	**296** (21.3)	456.33 (18.2)	318.6 (39.6)	495.33 (73.3)	**271.3** (18.4)	448.8 (25.14)	315.3 (25.82)	360 (77.4)
	$\mathrm{P_{OA}}$	**99.72** (0.09)	99.31 (0.30)	99.41 (0.41)	96.53 (1.06)	**99.76** (0.04)	99.25 (0.32)	99.69 (0.19)	97.72 (1.13)
	$\mathrm{P_{AA}}$	**99.38** (0.12)	99.22 (0.14)	99.23 (0.23	96.88 (1.04)	**99.42** (0.07)	99.17 (0.20)	99.42 (0.11)	98.14 (0.68)
Salinas	$n_{\mathrm{ulabeled}}^{\mathrm{total}}$	**173.3** (28.2)	365.8 (81.8)	198 (41.8)	242 (81.34)	**157.3** (34.1)	266.6 (55.9)	158.7 (21.65)	344.66 (92.1)
	$\mathrm{P_{OA}}$	99.88 (0.12)	99.73 (0.52)	**99.92** (0.02)	98.83 (0.44)	99.90 (0.06)	99.89 (0.08)	**99.93** (0.01)	99.44 (0.23)
	$\mathrm{P_{AA}}$	99.87 (0.08)	99.82 (0.23)	**99.88** (0.03)	98.95 (0.43)	**99.87** (0.05)	99.91 (0.04)	**99.87** (0.02)	99.53 (0.14)
U. Pavia	$n_{\mathrm{ulabeled}}^{\mathrm{total}}$	**218.6** (36.6)	238.2 (40.23)	281.3 (56.55)	282.6 (103.2)	**178** (18.97)	187.6 (40.02)	212.6 (28.14)	421.3 (116.9)
	$\mathrm{P_{OA}}$	**99.80** (0.12)	98.89 (0.16)	99.54 (0.27)	97.36 (0.84)	**99.86** (0.04)	99.80 (0.10)	99.82 (0.05)	99.11 (0.26)
	$\mathrm{P_{AA}}$	**99.61** (0.11)	99.06 (0.85)	99.42 (0.22)	97.30 (0.81)	**99.68** (0.09)	99.63 (0.12)	99.61 (0.10)	98.99 (0.27)

IRTSA/IFTSA-AL. However, it should be noted that SVM and ELM produced discrete maps but not probability maps. To address this issue, probabilistic SVM (PSVM) [36], [37] and multinomial logistic ELM (MLELM) [39] were used to estimate *a posteriori* probabilities $p(\delta(\mathbf{r}_n) = m|\mathbf{r}_n)$ as to compute the BT scores. Despite that many other probabilistic classifiers could also be used, SVM and ELM were selected because they have been widely used in HSIC.

To further conduct a fair comparative analysis on the classification performance of all test classifiers using fewer training samples, three initial labeled samples were selected for each class. In this case, the total number of labeled samples would be 48 for Indian Pines and Salinas and 27 for University of Pavia. Fig. 15 plots the P_{OA} curves versus number of training samples using IRTSA/IFTSA-AL coupled with four different classifiers, MLR, ELM, SVM, and KNN.

It is clearly shown in Fig. 15 that KNN was the worst classifier and MLR and ELM were two best classifiers where MLR produced higher accuracies for Indian Pines and Salinas but ELM performed slightly better than MLR for University of Pavia. In addition, SVM was not as effective as MLR and ELM in terms of P_{OA}. However, as more spatial classification information was included and new training samples were augmented via iterative processes, SVM began competitive and could achieve the accuracies close to that produced by MLR/ELM. On the whole, the experimental results showed that IRTSA/IFTSA-AL could always achieve high classification accuracy as long as an appropriate classifier was selected. In particular, it could make as simple as a classifier such as SVM more competitive as the iterative process begins to feed back more spatial classification information, while in the meantime also augmenting and updating training samples.

F. Discussion

Based on the above experiments, several discussions are noteworthy.

1) All the experimental results showed that IRTSA/IFTSA-AL using both feedback processes to expand data cubes and augment training sample

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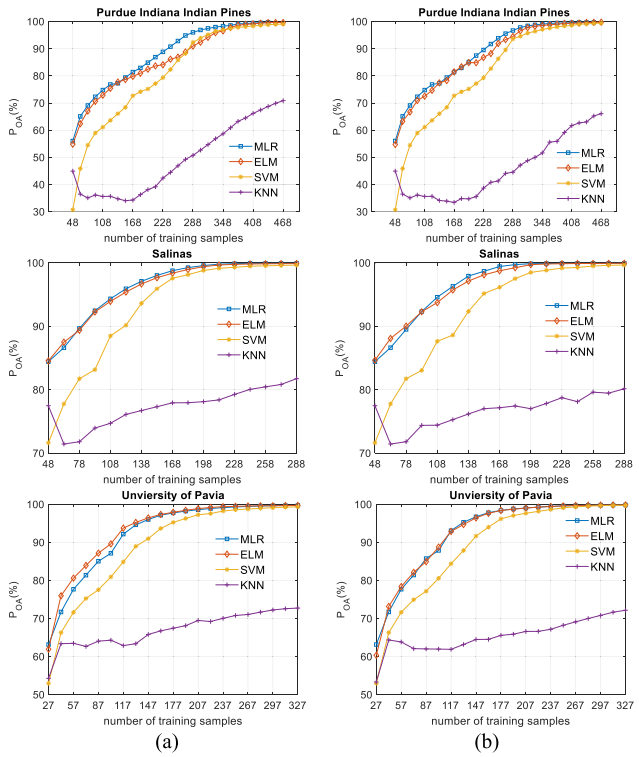


Fig. 15. P_{OA} of IFTSA-AL and IRTSA-AL using four spectral classifiers versus training sample number for three HSI data sets. (a) IFTSA-AL. (b) IRTSA-AL.

sets performed better than and were superior to AL-SS without the feedback process and IRTS/IFTS-SS without using AL criteria to augment training sample sets, specifically when the number of labeled training samples is small.

2) The training samples augmented by IFTSA-AL-SS can be dramatically different than those augmented by AL-SS, even though they both use the same initial set of labeled training samples, S_{labeled}, and the same identical probabilistic classifier to estimate *a posteriori* probabilities. This is mainly because AL-SS selects informative unlabeled samples by measuring the informativeness score on Ω_{HSI}. In contrast, the training samples augmented by IRTSA/IFTSA-AL are selected by uncertainty scores calculated for classified data samples on expanded data cubes, $\{\Omega^{(k)}\}_k$. More specifically, the augmented training samples obtained from $\Omega^{(k)}$ indeed include additional *a posteriori* spatial information that is fed back after each iteration, so that the reliability of *posteriori* probability estimation is improved and allows AL to select better training samples.
3) IRTSA/IFTSA-AL-SS can be considered as a generalization of ITS-SS and AL-SS. One drawback of IRTS/IFTS-SS is the computational time which is much longer than an SS method without implementing IRTS/IFTS due to repeatedly implementing feedback processes. Interestingly, IFTSA-AL-SS does not suffer from such issue because the AL and ITS can be implemented in parallel and simultaneously. When one process is terminated, the other is also terminated. As shown by experiments, ITS-AL-SS did not increase much more time than the AL-SS methods did.
4) As expected, IRTSA-AL-SS works better than IFTSA-AL-SS by including IRTS to update $S_{\mathrm{labeled}}^{(k)}$ in each iteration. The idea of IRTS was first developed and implemented with an SS classifier in [28], called IRTS-SS which has shown to outperform the IFTS-SS classifier. IRTSA-AL-SS not only augments unlabeled training samples but also reselects new set of labeled training samples in an iterative manner.

VII. Conclusion

This article presents a novel framework of AL-SS by combining the traditional AL-SS with the recently developed ITS-based methods, IRTS/IFTS-SS, to derive a new promising IRTSA/IFTSA-SS, which has the following unique features.

First, IRTSA/IFTSA-AL-SS combines both the strengths and advantages of IRTS/IFTS-SS and AL-SS to improve classification. Second, due to the use of the expanded data cube by IRTS/IT'S, the training sample augmentation by AL can produce better unlabeled training samples than AL-SS can. Third, IRTS/IFTS can be benefited from AL by augmenting training samples to further increase the already improved classification accuracies. Although IRTS/IFTS incorporates with AL, the experimental results demonstrated that IRTSA/IFTSA-AL-SS would not increase the computational time significantly compared with AL-SS. This is a great advantage because we can improve the performance of AL-SS without requiring much additional computing time. Fourth, by virtue of IRTS/IFTS, IRTSA/IFTSA-AL-SS can address the classification inconsistency caused by augmented training samples generated at each iteration. In particular, IRTSA-AL-SS has also shown to be more effective than IFTSA-AL-SS even if IFTSA-AL-SS is already performing very well. Fifth, the automatic stopping rule implemented in IRTS/ITS can also be applied to terminating the iterative process of training sample augmentation implemented in AL-SS. This is particularly important for practical applications to avoid the oversampling issue and reduce the labeling cost. Finally and most importantly, IRTSA/IFTSA-AL-SS does not specify any spectral classifier or SFs. As a matter of fact, any of the existing AL-SS methods can be extended to its IRTSA/IFTSA-AL-SS counterpart to further improve its classification performance. The experimental results of three benchmark hyperspectral image scenes clearly demonstrate their superiority to AL-SS and IRTS/IFTS-SS implemented alone in terms of P_{OA}, P_{AA}, and P_{OPR}.

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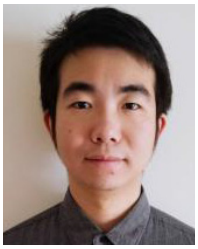
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